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# TOWARD COHERENCE DEPTH TWO FOR CUBICAL SEALED EXTENSIONS

HALVOR LANDE

Independent researcher, Oslo, Norway  
*e-mail address:* halvor.s.lande@gmail.com

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**ABSTRACT.** Type-theoretic foundations can grow by sealing globally acting structural layers: modalities, universe formers, promoted interface packages, and similar operations whose behavior becomes part of the public core. Such a layer must explain how it interacts with already exported structure; we call this residue its public structural trace. This paper proves a depth-two theorem for a fixed cubical sealed-extension calculus. Unary trace is not enough, because univalence turns equivalences such as the nontrivial swap of the two-point type into public paths. Higher structural trace generated by sealing is not primitive: it is represented by cubical open-box extension data computed from lower-arity boundaries. Consequently the exact realized obligation objects stabilize at binary trace, and normalized minimal public signatures contain no primitive structural trace field above binary arity. Separate full-coupling hypotheses give a two-step accounting recurrence; the Fibonacci form is only a constant-payload envelope. The mechanized development checks the theorem-facing fixed-calculus stack, the concrete SMod adequacy instance, semantic wrapper interfaces, case-study fixtures, and the theorem-map/postulate audit; broader transfer to other cubical foundations remains conditional on the stated interface.

## 1. INTRODUCTION

A type-theoretic library can grow in two very different ways. In the first, a user defines a new transparent object inside an already fixed foundation. A new datatype of trees, for example, need not explain how it interacts with every old construction in the library before it can be used. Its dependencies are local, and the surrounding theory can mostly ignore it.

The second kind of growth is less forgiving. A foundation can be extended by a new sealed structural layer: a modality, a universe former, a promoted interface package, or some other globally acting operation whose public behavior becomes part of the exported core. Such an extension is not merely another transparent term. Once sealed, it advertises a stable public interface. It must therefore explain how it acts on the active structure already present: old type formers, transport operations, eliminators, computation rules, paths, equivalences, and comparison data. This explanatory residue is what we call *public trace*.

The question of this paper is:

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When a cubical type-theoretic foundation is extended by a sealed, globally acting structural layer, how much public integration trace must remain primitive after normalization and opacity?

We keep one small running example in view. The old interface exports the two-point type  $2$  and the univalent swap path  $p_{\text{swap}} = \text{ua}(\text{swap}) : 2 = 2$ . A sealed operation  $X$  supplies an endomap  $F_2 : 2 \rightarrow 2$ . The constant-left choice is valid unary data but fails the binary comparison along  $p_{\text{swap}}$ ; the identity choice is the positive control. Later, after such binary boundary data is present, an additional generated compatibility becomes a structural horn whose missing face is supplied by the open-box construction.

The answer proved here is a depth-two theorem for one fixed formal calculus. In that calculus, unary trace is not enough: univalence supplies genuine binary naturality obligations. But trace above binary arity is not primitive: the higher structural obligations generated by sealing are cubical open-box extension problems, and cubical composition and filling compute their missing faces and fillers. Thus higher exact obligation factors remain visible in the semantics, but they disappear from minimal opaque public signatures.

A useful slogan is:

Binary trace is necessary; higher structural trace is cubically payable.

This slogan is deliberately about *structural integration trace*, not about all higher homotopy in a cubical theory. The theorem does not say that higher paths vanish, that user-supplied higher payload is forbidden, or that ordinary libraries grow according to a fixed recurrence. It says that, for the sealed extension calculus studied below, the additional public trace forced by sealing a globally acting structural layer stabilizes at binary trace.

**1.1. Payload, trace, and the running example.** The first distinction is between *payload* and *trace*. The payload is what the new sealed layer contributes: a new operation, modality, universe rule, or promoted structural package. The trace is the public evidence that this payload interacts correctly with the already exported interface.

Consider a small old interface containing two objects  $A$  and  $B$ , together with a comparison

$$p : A \simeq B$$

or, in a univalent presentation, a path obtained from such an equivalence. Now seal a new globally acting operation  $X$ . There are several levels of public obligation.

First, there is unary trace: the extension must say how  $X$  acts on individual old generators such as  $A$  and  $B$ . This is objectwise information. Second, there is binary trace: the extension must say that the action of  $X$  respects old comparison data such as  $p$ . This is not reducible to two unrelated unary actions, because  $p$  is part of the old public interface. A sealed action that acts correctly on  $A$  and  $B$  separately may still fail to commute with the public equivalence between them.

The running lower-bound example specializes this distinction. Univalence turns the boolean swap equivalence into a public path, and transport in the endomorphism family along that path conjugates an endomap by the swap. The identity endomap is fixed, while the constant-left endomap is sent to the constant-right endomap. Thus the unary fragment can accept objectwise data whose binary sealing obligation is uninhabited. Binary trace is a real public obligation, not an accounting artifact.

This is the first half of the depth-two story:

depth one is insufficient.

The constant-left branch proves this lower bound. The identity branch shows that binary trace is not an impossible demand, and it is the branch to keep in mind when the later open-box argument computes additional generated compatibilities from an already supplied binary boundary. The second half explains why the story stops at two.

**1.2. Why higher structural obligations are different.** At first sight, sealing seems to create an infinite tower. If unary action must respect binary comparison data, should binary comparison data itself not satisfy ternary compatibility laws? Should those laws not satisfy quaternary laws, and so on? This is the familiar shape of coherence problems.

Cubical foundations change the accounting. Their primitive operations are built to solve open-box extension problems. Given a partial boundary of a cube and a compatible base, cubical composition and filling supply the missing face and the interior filler. In a CCHM-style core [CCHM18] this is the role of the Kan operations, together with transport and the interval structure. The point of the present paper is that the higher structural obligations generated by sealing have exactly this open-box form.

The relevant higher obligation is not an arbitrary closed higher-dimensional choice. It is a *marginal extension problem*. The lower-dimensional faces come from the already exported unary and binary trace, together with normalized public signatures of previous sealed layers. What appears as a ternary or higher coherence field is therefore represented as an open box: a partial boundary, a compatibility condition, a missing remote face, and a filler. In the fixed calculus, this open-box extension object is contractible and substitution-stable. Its canonical point is computed using cubical composition and filling.

This is the key semantic mechanism. Higher structural obligations are not ignored. They remain present in the exact realized obligation object. But they are no longer primitive public data, because the calculus computes them from the lower-arity boundary. Equivalently, an explicit higher structural field in a surface presentation can be replaced by a cubical term built from the displayed boundary, and then erased from the minimal public signature.

In the running example, the constant-left branch has already failed before this stage. On the swap-respecting branch, the binary comparison is part of the public boundary; a later generated compatibility asks whether that lower boundary extends to a marginal box, not for an unrelated primitive field.

Thus the main theorem has two levels. At the level of exact realized obligation objects, higher factors are present but contractible. At the level of normalized opaque public signatures, higher structural fields are absent from the primitive trace count.

**1.3. The fixed calculus proved here.** Let  $C_{\text{ext}}$  be the sealed-extension calculus defined in this paper. It consists of a CCHM-style cubical core with univalence and Kan composition [CCHM18], globally acting extension declarations, opaque public signatures, and a raw adequacy bridge connecting surface presentations to normalized trace objects. The definitions of  $C_{\text{ext}}$  are part of the theorem object.

For an admissible candidate  $X$ , write  $\text{Real}_k(X)$  for the exact realized structural obligation object obtained when public trace is allowed up to arity  $k$ . Write  $\mu$  for the primitive public trace cost of a normalized opaque signature. The central result is:

**Main theorem for  $C_{\text{ext}}$ .** For every admissible candidate  $X$  and every  $k \geq 2$ , the restriction map

$$\text{res}_{2,k}(X) : \text{Real}_k(X) \longrightarrow \text{Real}_2(X)$$

is an equivalence. Its inverse  $\text{ext}_{2,k}(X)$  is constructed by adjoining, in dependency order, the canonical total open-extension witnesses for the higher structural obligations generated above binary arity. Moreover, every  $\mu$ -minimal normalized public signature contains no primitive structural trace field of arity greater than two. Unary trace is insufficient in general, as witnessed by the two-point swap obstruction.

The theorem should be read as an exact-depth statement for a fixed calculus:

binary trace is necessary,

and binary trace is sufficient for primitive public structural integration.

**1.4. Scope and non-goals.** The theorem is intentionally scoped. It is a theorem about the fixed calculus  $C_{\text{ext}}$ , not a claim that this is the only possible formalization of sealed cubical extension, nor that every Cubical Agda program [VMA19] automatically interprets into it. The result also does not say that all higher filler spaces in Homotopy Type Theory are contractible. The contractible objects are the theorem-facing structural open-box extension fibers generated by this calculus. User-supplied higher payload, raw higher homotopy, and non-structural mathematical data remain outside the primitive-trace accounting unless they are exported as sealing-generated structural integration fields.

The Fibonacci recurrence is scoped in the same way. It is an asymptotic worst-case envelope obtained only after adding recent-history factorization, full coupling, bundled per-site counting, uniform payload size, and the empty-history bootstrap. Sparse footprints and ordinary transparent library growth are expected to remain below that envelope.

The sufficiency direction is proved by the structural horn-to-open-box theorem, the open-box contractibility theorem, and the computational replacement theorem for explicit higher structural fields. The necessity direction is proved by the univalent boolean-swap obstruction.

The transfer interface isolates the corresponding adequacy criterion: a surface calculus or cubical foundation inherits the theorem when it admits a faithful interpretation preserving semantic admissibility, primitive-versus-derived trace status, historical support, horn-computational replacement, and public trace cardinality.

**1.5. What the paper proves, and what remains conditional.** The contribution of the paper is a theorem stack for the fixed calculus  $C_{\text{ext}}$ , together with a transfer interface for broader systems.

First, the paper defines a raw sealed-extension calculus that separates payload from public trace, primitive fields from derived horn-computational fields, exact realized obligation objects from  $\mu$ -minimal public signatures, and arity depth from chronological support.

Second, it proves that every higher structural obligation generated by sealing is represented as a theorem-facing open-box extension object. These open-box extension fibers are contractible and substitution-stable in the fixed cubical core. This is the semantic reason that higher exact factors do not contribute primitive public trace.

Third, it proves a computational replacement theorem. An explicit higher structural field in a presentation elaborates to a cubical term, built from composition, filling, and transport, using only its lower-arity boundary data. After normalization, the explicit field is absent from the  $\mu$ -minimal public signature.

Fourth, it proves exact structural integration depth two for  $C_{\text{ext}}$ :  $\text{Real}_k(X) \simeq \text{Real}_2(X)$  for  $k \geq 2$ , with no primitive public trace above binary arity, while the swap obstruction shows that unary trace alone is insufficient.

Fifth, it records downstream counting corollaries: sparse footprint accounting, the fully coupled affine recurrence, and the shifted Fibonacci specialization, all under the accounting hypotheses stated in Section 6.

Finally, the accompanying Cubical Agda development checks the theorem-facing fixed model stack: raw normalization and trace accounting, structural horn decoding, the open-box and horn-computation packages, computational replacement, wrappers for exact depth and chronological windows, lower-bound witnesses, 2D-foundation wrappers, recurrence arithmetic, semantic scaling wrappers, the concrete SMod surface-adequacy instance, and case-study audit fixtures. The aggregate artifact also checks the machine-readable theorem map and reports no local postulates or local primitive declarations in the theorem-facing import closure. The development is evidence for the fixed theorem and for the adequacy program, but the broad transfer claim remains conditional. In particular, this paper does not provide a parser or elaborator theorem for arbitrary Cubical Agda programs, nor a theorem for every cubical foundation.

**1.6. Related work.** The cubical core used here follows the line of cubical type theory initiated by Cohen, Coquand, Huber, and Mörtberg [CCHM18], including the use of composition, filling, and transport to give computational content to univalence. The paper also uses the implementation perspective of Cubical Agda [VMA19] as a nearby reference point, while keeping the theorem target smaller than the full surface language. Cartesian and other cubical systems [ABC<sup>+</sup>21, SAG22] provide additional context for how Kan structure and equality computation may be packaged.

The naive infinite-tower intuition comes from classical coherence phenomena: weak higher categories and weak  $\omega$ -groupoids contain coherence data at all dimensions, and identity types in intensional type theory carry weak  $\omega$ -groupoid structure [Lum10, vdBG11]. Classical coherence results for associativity and higher homotopy coherence also shape that expectation [Lan63, Sta63a, Sta63b, KvR20]. The present paper does not remove higher structure; it identifies a restricted class of sealing-generated structural obligations whose higher layers are open-box extension fibers.

Two-level type theory and related strict-equality extensions offer a different way to control coherence: they add a strict equality layer alongside homotopical equality [ACK16, ACKS23]. That approach can sidestep some higher coherence obligations by moving them to the strict layer. The collapse here is instead cubical: binary trace remains genuinely necessary in the univalent layer, while higher sealing-generated structural trace is computed from lower boundary data.

The opacity and export discipline is indebted to the module-system tradition. ML-style signatures, functors, sharing constraints, manifest types, and higher-order modules study how components expose stable interfaces while hiding implementation details [Mac84, HL94, Ler94, DCH03]. For proof assistants, module calculi for Pure Type Systems and Coq-style module implementations make the same modularity issues theorem-facing rather than purely programming-language-facing [Cou07, Chr04, FL04]. The sealed layers in this paper should be read against that background: the novel question is not whether abstraction barriers matter, but how much cubical structural trace such barriers force into the public interface.

**1.7. Roadmap.** Section 2 defines the sealed-extension calculus, the payload/trace distinction, primitive and derived public fields, and the trace-cost invariant  $\mu$ . Section 3 develops normalized public signatures and the raw adequacy bridge, and Section 4 isolates structural trace roles. Section 5 proves the open-box upper bound, replacement theorem, stabilization theorem, recent-history result, and swap lower bound. Sections 6 and 7 record the conditional counting corollary and the mechanized trusted boundary, and Section 8 concludes.

## 2. FIXED SEALED-EXTENSION CALCULUS

The preceding section stated the theorem informally: a sealed cubical extension must pay primitive public trace through binary comparison, but higher structural trace is computed by cubical horn filling. This section defines the fixed raw sealed-extension calculus  $C_{\text{ext}}$ , in which payload, trace, opacity, support, and trace cost can be measured without depending on a particular surface presentation.

**Fixed-calculus contract.** All depth and trace theorems in this paper concern the calculus  $C_{\text{ext}}$  defined here. The calculus consists of: (i) a cubical core with interval, cofibrations, transport, composition, filling, and univalence; (ii) raw sealed-extension declarations; (iii) public-signature extraction and normalization; (iv) a generated structural-obligation judgment; (v) depth-indexed obligation signatures  $\text{ObSig}_k$ ; (vi) realization objects  $\text{Real}_k$  with restriction maps; and (vii) a primitive/derived trace classification assigned only after explicit replacement witnesses have been constructed. The depth-two theorem is not a claim about arbitrary Cubical Agda developments or arbitrary cubical foundations without an adequacy package.

The section has three jobs. First, it explains what counts as a sealed layer and what is visible after sealing. Second, it defines the normalized public signatures on which all cardinality statements are made. Third, it isolates the special fully coupled regime used by the recurrence theorem. The depth-two theorem itself is not proved here: the horn-computing argument belongs to Sections 4 and 5. What this section supplies is the accounting language in which that theorem can be stated.

**2.1. Notation and terminology.** A library state is written  $B$ . Its sealed layers are written  $L_0, L_1, \dots$ , and  $S(L_i)$  denotes the atomic public signature exported by  $L_i$  after counting normalization. A new declaration is written

$$e = (\Gamma_{\text{new}}, \Gamma_{\text{act}}, \Gamma_{\text{cmp}}, \sigma).$$

The component  $\Gamma_{\text{new}}$  contains payload,  $\Gamma_{\text{act}}$  contains unary action trace,  $\Gamma_{\text{cmp}}$  contains comparison and horn-boundary trace, and  $\sigma$  selects what is exported through the opaque public interface.

The following vocabulary is used throughout the paper.

| Term                 | Meaning in this paper                                                               |
|----------------------|-------------------------------------------------------------------------------------|
| Sealed extension     | A new opaque layer added to the exported foundation or public interface.            |
| Payload              | The genuinely new mathematical or operational structure supplied by the layer.      |
| Trace                | Public compatibility data required for the payload to interact with prior exports.  |
| Primitive trace      | Trace not synthesized by normalization or cubical structural computation.           |
| Derived trace        | Trace present in a presentation but computed from lower data or generic operations. |
| Counting normal form | The canonical flattened telescope on which public fields are counted.               |
| Active basis site    | One irreducible primitive class of an earlier normalized public signature.          |
| Full coupling        | The regime in which the new layer acts on every active basis site in scope.         |
| Sparse footprint     | The general regime in which only selected active sites are touched.                 |
| Adequacy boundary    | The bridge from richer semantic or surface systems into this raw calculus.          |

Two separations should be kept in mind. First, payload may be higher-dimensional. A user may add higher inductive constructors, path constructors, or other genuinely higher content. Those are counted as payload when they are part of the new layer's mathematical content. Only the compatibility data forced by sealing the new layer against old public interfaces is called structural trace. Second, exact realized obligation objects and minimal public signatures are different. A higher horn obligation may remain present as a contractible exact factor while contributing no primitive field to the minimal opaque signature.

**2.2. Raw syntax and judgments.** The theorem target is the following raw extension calculus. Its grammar is deliberately small:

$$\begin{aligned}
& i, j, n \in \text{LayerId} \\
B & ::= \cdot \mid B, L_i, \\
e & ::= (\Gamma_{\text{new}}, \Gamma_{\text{act}}, \Gamma_{\text{cmp}}, \sigma), \\
\phi & ::= \text{Payload}(q) \mid \text{Act}(s) \mid \text{Cmp}(s, t, p) \mid \text{Horn}_k(\partial), \\
\Sigma & ::= \cdot \mid \Sigma, \phi : T.
\end{aligned}$$

Here  $B$  is a monotone state of sealed layers. The new declaration  $e$  has four parts. The payload telescope  $\Gamma_{\text{new}}$  introduces the new symbols of the layer. The action telescope  $\Gamma_{\text{act}}$  describes unary action on selected old interface sites. The comparison telescope  $\Gamma_{\text{cmp}}$  describes binary comparison, transport, naturality, and horn-boundary data. The export selection  $\sigma$  records which elaborated fields are visible after sealing. The field grammar separates payload from structural trace:  $\text{Payload}(q)$  belongs to the new mathematical content, while  $\text{Act}$ ,  $\text{Cmp}$ , and  $\text{Horn}$  are the theorem-facing structural trace shapes.

$$\begin{array}{c}
\frac{B \text{ wf} \quad s \in I(B) \quad B, \Gamma_{\text{new}} \vdash a_s : \mathbf{Act}(X, s)}{B \vdash a_s : \mathbf{Unary}(X; s)} \text{ (ACT)} \\
\frac{B \vdash a_s : \mathbf{Unary}(X; s) \quad B \vdash a_t : \mathbf{Unary}(X; t) \quad p : s \simeq t \in I(B)}{B \vdash b_p : \mathbf{Cmp}(a_s, a_t, p)} \text{ (CMP)} \\
\frac{B \vdash b_p : \mathbf{Cmp}(a_s, a_t, p)}{B \vdash b_p : \mathbf{Binary}(X; s, t, p)} \\
\frac{B \vdash \partial : \mathbf{Boundary}_k(X) \quad k \geq 3}{B \vdash h : \mathbf{OpenExt}_k(\partial)} \text{ (HORN)} \\
\frac{B \vdash h : \mathbf{OpenExt}_k(\partial)}{B \vdash h : \mathbf{Horn}_k(X; \partial)} \\
\frac{B \vdash e : \mathbf{RawDecl} \quad B \vdash e \text{ adm} \quad B \vdash N(e) : \mathbf{NormSig}}{B \vdash \text{Seal}(e) : \mathbf{Layer} \quad B, \text{Seal}(e) \text{ wf}} \text{ (SEAL)}
\end{array}$$

Figure 1: Core theorem-facing rules of the sealed-extension calculus. The judgments use the schematic families **Act**, **Cmp**, and **Boundary**<sub>k</sub> defined by the raw declaration being elaborated; the later sections make their normalization and derivedness criteria precise.

The fixed calculus is specified by the following judgments:

|                                                                   |                                               |
|-------------------------------------------------------------------|-----------------------------------------------|
| $B \text{ wf}$                                                    | well-formed states,                           |
| $B \vdash e : \mathbf{RawDecl}$                                   | raw declarations,                             |
| $B \vdash e \text{ adm}$                                          | admissible declarations,                      |
| $B \vdash \text{Elab}(e) : \mathbf{Cand}$                         | elaboration to candidates,                    |
| $B \vdash \text{Seal}(e) : \mathbf{Layer}$                        | opaque sealing,                               |
| $B \vdash N(e) : \mathbf{NormSig}$                                | normalized public signatures,                 |
| $B \vdash \phi : \mathbf{RawField}$                               | selected public fields before classification, |
| $B \vdash \text{class}(\phi) \in \{\mathbf{Prim}, \mathbf{Der}\}$ | post-replacement class,                       |
| $B \vdash \text{Supp}(\phi) = S$                                  | historical support.                           |

An admissible declaration is therefore not just a syntactic tuple. Its payload, action clauses, comparison clauses, horn-boundary clauses, and export selection must type-check; the selected footprint must match the advertised coupling mode; and the exported interface must normalize to an opaque public signature.

**Definition 2.1** (Generated structural obligations). For a cubical context  $\Gamma$ , a normalized old public signature  $\Sigma$ , and a candidate  $X$ , the structural obligations generated by sealing  $X$  over  $\Sigma$  are the least arity-indexed family of judgments

$$\Gamma; \Sigma \vdash \tau \in \mathbf{SObl}_n(X)$$

closed under the ACT, CMP, and HORN constructors of Figure 1, and stable under weakening, substitution, and presentation equivalence of  $\Sigma$ . The arity  $n$  is the maximum public boundary dimension inspected by  $\tau$ : action obligations have arity one, comparison obligations have arity two, and structural horn obligations have arity  $n \geq 3$ . Payload declarations are not members of  $\mathbf{SObl}_n(X)$  unless the export selection explicitly reclassifies them as structural integration trace.



**Definition 2.2** (Depth-indexed admissibility). For  $k \in \mathbb{N} \cup \{\infty\}$ , a candidate  $X$  is  $k$ -admissible when every generated obligation  $\tau \in \text{SObl}_n(X)$  with  $n \leq k$  has evidence in the corresponding realized obligation object. Unary admissibility means 1-admissibility, binary admissibility means 2-admissibility, and full admissibility means  $\infty$ -admissibility. The paper does not define full admissibility by appealing to the depth-two theorem; the theorem later proves that, for  $C_{\text{ext}}$ , binary admissibility extends canonically to full structural admissibility.

**Definition 2.3** (Raw structural clause classes). The raw calculus uses three theorem-facing structural clause classes.

A *raw action clause* is a field of  $\Gamma_{\text{act}}$  whose support mentions one prior opaque interface site and specifies the new payload’s unary action on that site.

A *raw comparison clause* is a field of  $\Gamma_{\text{cmp}}$  whose support mentions two prior interface sites and compares the two induced unary composites. These include the transport and naturality squares forced by old public paths or equivalences.

A *raw horn clause* is a field of  $\Gamma_{\text{cmp}}$  whose type presents a higher structural boundary together with the missing-face and filler package of Section 5. Raw horn clauses may occur in exact realized obligation objects. After the adequacy bridge and the computational replacement theorem, they are classified as derived trace exactly when their witness is synthesized by the cubical Kan operations from the lower boundary.

The last sentence is important. The word “derived” is not meant to be a mere syntactic tag. The raw syntax records where a horn-shaped structural field appears; Sections 4 and 5 prove that the relevant fields are computed by explicit cubical operations and therefore disappear from  $\mu$ -minimal public trace.

### 2.3. States, extensions, and public signatures.

**Definition 2.4** (State). A *state* or *library* is a monotone context  $B$  closed under derivability. An evolution step

$$B_n \rightsquigarrow B_{n+1}$$

adjoins one sealed integration layer, denoted  $L_{n+1}$ .

A state is not the same thing as a source file or an informal collection of lemmas. It is the opaque public interface available to future sealed extensions. Transparent internal construction may be used to build a layer, but after sealing only the selected normalized fields of that layer are visible.

**Definition 2.5** (Surface extensions). Over a state  $B_n$ , a surface extension declaration is a finite quadruple

$$e = (\Gamma_{\text{new}}, \Gamma_{\text{act}}, \Gamma_{\text{cmp}}, \sigma),$$

where  $\Gamma_{\text{new}}$  is a telescope of newly declared payload symbols,  $\Gamma_{\text{act}}$  is a finite family of unary action clauses against the active interface,  $\Gamma_{\text{cmp}}$  is a finite family of comparison, transport, or horn-boundary clauses, and  $\sigma$  specifies which elaborated fields are exported through the sealed public interface. The admissible extensions studied in this paper are exactly those declarations whose intended action is global on the designated active interface footprint.

**Definition 2.6** (Elaboration, sealing, and normalized public signature). Each admissible surface extension declaration  $e$  over  $B_n$  elaborates to a core object

$$\text{Elab}(e)$$

of the fixed cubical core  $C$ . The payload telescope elaborates to new core declarations. The action and comparison clauses elaborate to fields witnessing interaction with the active interface. The opacity component  $\sigma$  selects the public telescope exported after sealing.

The *normalized public signature* of  $e$  is the counting normal form

$$N(e) := N(\text{Elab}(e)).$$

It is obtained by flattening the exported public telescope, quotienting transparent telescope isomorphisms, and recording the transparent synthesis candidates generated by definitional equality, reflexivity, transport along already exported equalities, generic structural operations of the ambient cubical calculus, and later derived-trace witnesses. The primitive/derived split is assigned only after the replacement theorem has supplied those witnesses.

**Definition 2.7** (Presentation equivalence and accounting notation). Two admissible surface extensions  $e$  and  $e'$  over the same state are *presentation-equivalent*, written

$$e \approx e',$$

when their elaborated sealed extensions determine the same public typing judgment up to definitional equality, canonical telescope isomorphism, and transparent internal normalization.

Once the replacement theorem has been applied, write  $N_{\text{tr}}^{\text{prim}}(e) \subseteq N(e)$  for the irreducible primitive trace schemas in the normalized public signature of  $e$ . The notation

$$\mu(e) := \min\{|N_{\text{tr}}^{\text{prim}}(e')| \mid e' \approx e\}$$

is reserved here for that later minimal opaque cost. In the mechanization, the admissible presentation moves are generated by explicit presentation steps, including reassociation, record splitting and bundling, currying/uncurrying, transport of fields, transparent alias insertion or deletion, and duplicate-derived deletion.

Definition 2.7 is only accounting notation. The formal well-definedness of the primitive part used by  $\mu$  is not taken from a raw grammar tag: it is justified by the computational replacement theorem of Section 3 and by the **DerivedTrace** witnesses constructed in Sections 4 and 5. Until those results have been proved, “primitive” and “derived” name the two cases of the later canonical normal form, not extra input data supplied by raw declarations.

The invariant  $\mu$  is deliberately insensitive to packaging. A trace square does not become cheaper by being hidden inside a one-constructor record, and it does not become more expensive by being presented as several equivalent telescope spines. What  $\mu$  counts is irreducible primitive public trace after canonical normalization.

**Definition 2.8** (Candidate). A *candidate* over a library state  $B$  is a pair

$$X = (X_{\text{core}}, G_{\text{obl}}),$$

where  $X_{\text{core}}$  is the mathematical content proposed for addition and  $G_{\text{obl}}$  is the family of normalized atomic structural obligations induced when  $X$  is sealed against the active interface of  $B$ . Every admissible surface extension declaration  $e$  determines such a candidate by elaboration.

**Definition 2.9** (Counting normal form and atomic schemas). Every public interface in this paper is counted only up to canonical telescope isomorphism: reassociation and flattening of iterated dependent sums, splitting or rebundling a single-constructor record into nested

records, and currying or uncurrying iterated function arguments when these preserve the same sealed typing judgment. Concretely, an exported interface is read as a dependent record, equivalently an iterated  $\Sigma$ -type whose field types may themselves begin with  $\Pi$ -telescopes. Its *counting normal form* is the fully flattened telescope obtained by reassociating the  $\Sigma$ -spine and expanding such record wrappers.

An *atomic schema* is one field of this counting normal form after transparent telescope isomorphisms have been quotiented. After the replacement theorem has been applied, such a field is classified as *derived* when its inhabitant is uniformly synthesized from earlier fields by definitional equality, reflexivity, transport along already exported equalities, generic structural operations of the ambient calculus, or a derived trace witness in the sense of Definition 4.10. It is classified as *primitive* when no such transparent synthesis is available in the fixed calculus. The cost and basis notions below count primitive atomic schemas after this later classification; derived schemas may remain present in exact signatures but contribute no primitive public trace unit.

**Definition 2.10** (Core payload size). For a candidate  $X$ , let  $P(X)$  be the finite set of atomic payload schemas appearing in the counting normal form of the public interface contributed directly by  $X_{\text{core}}$  before any structural witnesses against earlier layers are added. Its cardinality

$$\kappa(X) := |P(X)|$$

is the *core payload size* of  $X$ .

**Definition 2.11** (Sealed extension). A *sealed extension* is a candidate whose structural obligations have been discharged and whose resolved data are exported only through an opaque public interface. The opacity invariant is the rule

future layers may depend on  $S(L_i)$ , but not on the derivation of  $\text{Seal}(e_i)$ .

Thus future extensions may use exported payload and trace fields, but they may not inspect the internal derivations that produced them.

Opacity is what makes trace visible. If future layers could freely inspect every proof term and normalization derivation used to build  $L_i$ , then some apparent public trace could be recovered by peeking behind the interface. A sealed extension disallows that move: the normalized public signature is the contract with later layers.

**Definition 2.12** (Semantic sealed layer and public interface). At the semantic level the  $n$ th sealed layer may be read as a package

$$E_n := (B_n, K_n, T_n, \text{interp}_n).$$

Here  $B_n$  is the incoming library state,  $K_n$  is the payload telescope contributed by the new content,  $T_n$  is the resolved structural trace telescope exported after sealing, and  $\text{interp}_n$  interprets both parts in the ambient cubical foundation. The public interface of the sealed layer is the tagged coproduct

$$I_n := K_n \sqcup T_n.$$

For the fixed raw instance,  $K_n$  is the payload part  $S_{\text{core}}(L_n)$  of the normalized public signature and  $T_n$  is the trace part  $S_{\text{tr}}(L_n)$ . User-supplied higher constructors and other mathematical content belong to  $K_n$  even when they are genuinely higher-dimensional; only the integration data induced by sealing the layer against earlier opaque interfaces belongs to  $T_n$ .

**Definition 2.13** (Promoted interface package). A *promoted interface package* over  $B$  is a finite package of constructively irreducible generators and interaction data that is exported as one new sealed layer. Such a package need not modify the reduction rules of the kernel. What matters is that its public generators become part of the active exported interface for later sealing steps.

**Definition 2.14** (Integration latency). The *integration latency* of a candidate  $X$  over  $B$  is

$$\Delta(X \mid B) := |G_{\text{obl}}|,$$

the number of irreducible atomic structural obligations that must be resolved before  $X$  can be exported as a sealed layer.

**Remark 2.15** (Refactoring invariance of the counting notions). Because  $\kappa$ ,  $\Delta$ , and later historical support are computed on counting normal forms, admissible refactorings such as bundling laws into one record, splitting one interface into several record shells, exporting curried rather than uncurried arguments, or pushing transparently derivable witnesses deeper into an internal term do not change the counts. Only a change in the irreducible primitive fields of the sealed typing judgment changes the metrics.

For a sealed layer  $L_k$  obtained from a candidate  $X_k$ , we write  $S(L_k)$  for the set of atomic exported schemas in the counting normal form of its public interface and

$$\kappa_k := \kappa(X_k)$$

for its core payload size. The indexing is chosen so that  $\Delta_k$  denotes the sealing cost of  $L_k$ .

Transparent user-level code. A family  $U$  of transparent definitions, lemmas, or record packages over  $B$  lies outside the recurrence when its symbols are definitionally reducible using the existing operational semantics and are not promoted to a new opaque exported layer. Such a development contributes zero integration latency and does not alter the active interface seen by later sealing steps: all reductions remain inherited from the pre-existing library, and no new opaque generators are added to the global interface telescope. It is ordinary growth inside one state, not a state transition of the form  $B_n \rightsquigarrow B_{n+1}$ .

**Definition 2.16** (Foundational core extension). A candidate  $X$  over  $B_n$  is a *foundational core extension* if its payload package is intended to act globally on the currently active interface: the whole payload telescope  $P(X)$  is operationally or semantically incomplete until its behavior has been specified against every basis site in the designated coupling footprint inside the active exported interface. Typical examples are extensions of a universe-decoding apparatus, global modalities presented by explicit modal dependent-type syntax whose context action must distribute across existing type formers and transports, and heavily promoted generic interfaces whose exported action is library-wide.

**Remark 2.17.** Not every sealed extension is foundational in the sense of the previous definition. A new datatype or API layer may be orthogonal to much of the existing library and therefore require only sparse interaction data. Such extensions may preserve canonicity without becoming globally coupled to earlier layers. The recurrence theorem is intentionally a worst-case statement about foundational core extensions, not a universality claim about all sealed library growth.

This scope restriction is essential. Ordinary user-level additions typically enlarge a sparse DAG of modules, datatypes, or APIs and do not force primitive interaction with

unrelated earlier exports. The maximal-coupling results below isolate the special regime in which the new sealed layer is advertised as a global operation on the active interface. In that regime, coverage of the advertised footprint is an exhaustiveness requirement rather than a stylistic bookkeeping convention.

**2.4. Historical support, primitive classes, and bases.** The trace cost of a new layer depends not on the raw number of old fields but on the number of old primitive classes that remain visible after normalization. To make this invariant under presentation, we choose basis representatives only after quotienting by transparent derivability.

**Definition 2.18** (Historical interface). For  $k \geq 1$ , the *historical interface of depth  $k$*  at stage  $n$  is

$$I_n^{(k)} := \prod_{j=0}^{k-1} S(L_{n-j}),$$

the tagged coproduct of the atomic schemas exported by the most recent  $k$  sealed layers in counting normal form.

The tags matter: if two layers happen to export isomorphic-looking fields, they are still different public sites unless the normalized sealed interface identifies them.

**Definition 2.19** (Primitive classes and basis families). For each sealed layer  $L$ , let

$$\text{PrimClasses}(S(L))$$

be the finite set of primitive normal-form classes of the normalized public signature of  $L$ . Two primitive fields represent the same class exactly when counting normalization identifies them by transparent generation, definitional equality, uniform polymorphic instantiation, transport along already exported equalities, or the generic structural operations quotiented in Definition 2.9.

A *basis family*  $S_{\text{bas}}(L)$  is a chosen transversal of  $\text{PrimClasses}(S(L))$ , that is, one representative primitive field from each primitive normal-form class. For  $k \geq 1$ , once a basis family has been chosen for each of the layers  $L_n, \dots, L_{n-k+1}$ , the associated *historical basis of depth  $k$*  at stage  $n$  is

$$I_{n,\text{bas}}^{(k)} := \prod_{j=0}^{k-1} S_{\text{bas}}(L_{n-j}).$$

Its elements are the basis sites on which a globally acting extension is primitively specified.

**Theorem 2.20** (Canonical primitive classes determine bases). *For every sealed layer  $L$  in counting normal form, basis families in the sense of Definition 2.19 exist. If  $S_{\text{bas}}(L)$  and  $S'_{\text{bas}}(L)$  are two such basis families, then projection to primitive normal-form classes identifies a canonical bijection*

$$S_{\text{bas}}(L) \cong S'_{\text{bas}}(L).$$

*Consequently the cardinality  $|S_{\text{bas}}(L)|$  is exactly  $|\text{PrimClasses}(S(L))|$ , depends only on the canonical normalized public signature of  $L$ , and for every depth  $k$  the historical basis cardinality  $|I_{n,\text{bas}}^{(k)}|$  is independent of the chosen transversals and stable under presentation-equivalent replacement of the participating layers.*

The proof is by finite transversals of the primitive normal-form quotient. Each class has a representative, and two choices are canonically matched by projection to the same primitive class. Presentation-equivalent layers have canonically isomorphic normalized signatures by the normalization theorem of Section 3; this isomorphism preserves witnessed primitive/derived classifications and transparent-generation classes. Thus the invariant is the primitive normal-form class set itself, not an arbitrary minimal generating family.

**Primitive basis invariants.** For every sealed layer  $L$  and every presentation-equivalent replacement  $L'$ , the finite primitive class set, any basis-family cardinality, and the historical support and arity of every primitive trace field are canonically invariant. This follows by transporting along the canonical isomorphism of normalized public signatures: Theorem 2.20 identifies basis families with primitive normal-form class sets, and the normalization theorem of Section 3 preserves primitive/derived classifications, transparent-generation classes, and opaque layer identifiers.

**2.5. Global action, basis coverage, and canonical minimality.** This subsection isolates the hypothesis used by the recurrence theorem. It is not a theorem that every sensible extension is fully coupled. It is a theorem about extensions that advertise a total global action on a designated active footprint. Once such a footprint is fixed, coverage and minimality turn it into a precise trace count.

**Definition 2.21** (Basis coverage and canonical basis minimality). An admissible foundational core candidate sealed against  $I_n^{(k)}$  has *basis coverage* if, for any historical basis  $I_{n,\text{bas}}^{(k)}$  supplied by Definition 2.19, every basis site receives at least one primitive interaction datum from the new extension. It is in *canonical basis-minimal form* if, again for any such historical basis, after elaboration and normalization each basis site contributes at most one irreducible public trace schema. By Theorem 2.20, these properties do not depend on which basis family is chosen.

**Definition 2.22** (Active footprint and full coupling judgment). For a state  $B_n$  and a depth  $d$ , an *active footprint* is a finite subset

$$\text{Foot}_n^{(d)} \subseteq I_{n,\text{bas}}^{(d)}.$$

A declaration is *sparse at depth  $d$*  when its footprint is a proper subset of the historical basis. It is *fully coupled at depth  $d$* , written

$$B_n \vdash e \text{ fully-coupled}(d),$$

when its admissibility judgment includes primitive action and comparison slots, modulo canonical derivability, for every site of

$$\text{Foot}_{n,\text{full}}^{(d)} := I_{n,\text{bas}}^{(d)}.$$

Thus full coupling is an explicit hypothesis about the selected footprint, while sparse footprints remain available for ordinary local or orthogonal growth.

**Remark 2.23** (Trace-counting convention). The recurrence uses one trace schema per active basis site per sealed layer. A trace schema may be a dependent record packaging the behavior of the whole new payload telescope at that site; the individual clauses inside that package are not counted separately after bundling and counting normalization. A different convention, for example one primitive trace schema per payload generator per active basis site, would

insert an additional payload factor into the cardinality theorem below and would change the recurrence. This paper consistently uses the bundled per-site convention.

**Definition 2.24** (Fully coupled foundation). A foundation is *fully coupled* if the admissible sealed extensions under study are foundational core extensions satisfying the full-coupling judgment of Definition 2.22 at the relevant depth. Equivalently, the paper’s fully coupled declarations are globally acting payload packages whose intended coupling footprint is the whole active historical basis, not orthogonal additions with sparse dependency patterns. This definition fixes the scope of the action as a hypothesis. The coverage and minimality theorems below derive the exact irreducible primitive interaction data required to realize that action under the bundled per-site trace convention of Remark 2.23.

The next theorem turns basis coverage into a metatheoretic necessity for this globally coupled regime rather than a stylistic choice.

**Theorem 2.25** (Full-coupling coverage in the fixed native-totality record). *Let  $B_n$  be a library state in a fully coupled foundation, let  $X$  be an admissible foundational core extension sealed against the historical interface  $I_n^{(k)}$ , and let  $I_{n,\text{bas}}^{(k)}$  be any historical basis for that interface. If the extended theory is to pass the sealed library’s global admissibility discipline, then every basis site of  $I_{n,\text{bas}}^{(k)}$  must receive at least one primitive interaction datum from  $X$ . For native core extensions this is enforced by canonicity-preserving totality of the operational semantics; for promoted interface packages it is enforced by operational exhaustiveness of the exported global interface.*

*Proof.* By Definition 2.24, the admissible candidates considered here are not arbitrary sealed additions but foundational core extensions whose payload is intended to act on the whole active interface. The theorem is therefore about that global regime only; orthogonal extensions of the kind noted in Remark 2.17 are outside its hypotheses.

First consider a native global computational extension. In a Tarski-style universe presentation [ML84], suppose the active interface exports a code universe  $U$  and a decoding map  $\text{El} : U \rightarrow \text{Type}$ . If the new sealed layer extends the universe with a fresh code but the extended decoding machinery omits one basis site of the active interface, then by Definition 2.19 there remains a family of raw exported generators whose behavior is not recovered from the remaining primitive clauses by definitional equality, polymorphic instantiation, transport, or generic structural operations. Closed terms mentioning that uncovered family no longer reduce through the advertised global operator, so the extension is not total on the active interface it claims to cover. In the native case this appears as a canonicity or stuck-neutral failure for the extended language. Thus native admissibility forces at least one clause for every basis site of  $I_{n,\text{bas}}^{(k)}$ .

Now consider a promoted interface package. Such a package need not alter kernel reduction, so standard canonicity is not the right diagnostic. What matters instead is operational exhaustiveness of the exported global interface. If the package is advertised as a library-wide action, its public typing judgment is a total dependent map out of the record enumerating the active interface. Omitting one basis site leaves an irreducible family of cases uncovered, because that site’s action is not derivable from the other clauses by the transparent operations built into Definition 2.19. Later sealing steps cannot therefore treat the package as a closed global operation. Thus promoted packages also require at least one primitive witness for every basis site of  $I_{n,\text{bas}}^{(k)}$ .  $\square$

**Proposition 2.26** (Canonical normalized presentations are basis-minimal). *Let  $e$  be an admissible surface extension declaration over  $B_n$  whose candidate  $X(e)$  is sealed against the historical interface  $I_n^{(k)}$ , and let  $I_{n,\text{bas}}^{(k)}$  be any historical basis for that interface. Then the normalized public signature  $N(e)$  contains at most one irreducible public trace schema for each basis site of  $I_{n,\text{bas}}^{(k)}$ . Equivalently, redundant primitive clauses that differ only by transparent polymorphic compression, definitional equality, transport, or generic structural operations do not survive in the canonical normalized presentation.*

Indeed, two primitive clauses for the same basis site induce the same public trace field up to the definitional equalities, transparent transports, uniform polymorphic instantiations, and generic structural operations quotiented by counting normalization. If both survived in  $N(e)$ , the sealed public signature would still contain presentation-equivalent duplicate trace data, contrary to the canonical normalized presentation of Section 3. Hence at most one irreducible trace schema survives per basis site.

**Theorem 2.27** (Active-basis cardinality under full-coupling coverage). *Let  $B_n$  be a library state, let  $e$  be an admissible foundational core declaration over  $B_n$ , and suppose*

$$B_n \vdash e \text{ fully-coupled}(k)$$

*in the sense of Definition 2.22. Let  $X(e)$  be its elaborated candidate and let  $I_{n,\text{bas}}^{(k)}$  be any historical basis for the active interface. Under the bundled per-site convention of Remark 2.23,*

$$\Delta(X(e) \mid B_n) = |\text{Foot}_{n,\text{full}}^{(k)}| = |I_{n,\text{bas}}^{(k)}|.$$

*In particular, sealing cost counts irreducible basis clauses rather than the raw number of exported generators, and the right-hand side is independent of the chosen basis family by Theorem 2.20.*

The proof just combines the preceding coverage and minimality facts. The judgment  $B_n \vdash e \text{ fully-coupled}(k)$  identifies the active footprint with  $I_{n,\text{bas}}^{(k)}$ . Theorem 2.25 supplies at least one primitive interaction datum per basis site, and Proposition 2.26 leaves at most one irreducible schema per basis site in canonical normal form. Since  $\Delta(X(e) \mid B_n)$  counts those irreducible primitive obligations, the displayed cardinality identity follows.

### 3. THE ADEQUACY BOUNDARY AND NORMALIZED PUBLIC SIGNATURES

Section 2 defined the raw sealed-extension calculus and the accounting invariant  $\mu$ . Those definitions are intentionally presentation-insensitive: they count only the normalized opaque public signature exported by a sealed layer. This section explains how an extension declaration reaches that normal form. It is the place where surface syntax, semantic admissibility, opacity, and trace accounting meet.

The main point is a boundary rather than a slogan: all semantic readings pass through an *adequacy bridge*, a package of preservation claims connecting a surface or semantic extension language to the raw normalized signatures counted in Section 2.

The section has four jobs. First, it states the adequacy package that a surface language or external cubical foundation must supply in order to inherit the results of the paper. Second, it gives a small concrete instance, **SMod**, showing that the boundary is not empty. Third, it defines the presentation moves that normalization is allowed to forget. Finally, it



proves that the resulting normalized public signatures preserve exactly the trace information used later by the horn-computation, depth, and recurrence theorems.

A useful way to read this section is:

$$\begin{aligned} \text{surface declaration} &\rightsquigarrow \text{raw admissible declaration} \rightsquigarrow \text{sealed public interface} \\ &\rightsquigarrow \text{normalized public signature.} \end{aligned}$$

The depth theorem begins only after the last arrow. This is what prevents the later cardinality statements from depending on arbitrary choices of record packaging, telescope bracketing, or explicit versus implicit derived fields.

**3.1. Semantic obligations and the adequacy package.** A sealed extension is not counted merely by looking at the symbols a programmer writes. The same public interface may be presented using a record, an iterated dependent sum, a bundled operation, an unbundled telescope, or an explicit derived compatibility field. Conversely, small changes in surface notation may hide a real new primitive public obligation. The adequacy bridge identifies which fields are semantic payload, which fields are primitive public trace, and which fields are derived before  $\mu$  is computed.

**Definition 3.1** (Semantic structural obligations). Let  $E_{n+1}$  be a semantic sealed layer over an already sealed state  $B_n$ . A *semantic structural obligation* for  $E_{n+1}$  is an interface field required to interpret the new payload of  $E_{n+1}$  against the opaque public interfaces exported by previous layers.

Unary obligations specify the action of the new payload on one old public site. Binary obligations specify compatibility with an old comparison, path, transport, or equivalence between public sites. Higher horn-shaped obligations specify the remaining faces of a higher structural compatibility problem generated by sealing.

User-authored higher payload is not, by itself, structural trace. If a layer contributes a higher inductive constructor, a path constructor, or another genuinely higher mathematical operation, that data belongs to the payload part  $K_{n+1}$ . It becomes structural trace only when it is a public compatibility field forced by sealing the new layer against old opaque interfaces.

The last sentence is one of the most important discipline rules of the paper. The theorem is not claiming that higher-dimensional mathematical content disappears. It is claiming that higher *structural integration obligations* generated by sealing are represented as cubical horn-extension problems and therefore do not survive as primitive public trace.

**Remark 3.2** (Adequacy package and status). For a general cubical foundation  $\mathcal{F}$ , or for a richer surface language intended to compile into  $C_{\text{ext}}$ , the semantic reading of the theorems is parametric in an adequacy package  $\mathcal{A}$ . The package contains the following preservation clauses.

- (1) **Raw soundness.** Every well-typed surface payload, action, comparison, transport, or horn-boundary declaration elaborates to the corresponding raw clause class of  $C_{\text{ext}}$ .
- (2) **Raw completeness for the fixed sealing discipline.** Every admissible semantic structural obligation generated by the chosen sealing discipline is represented by one of the raw clause classes of Section 2: payload, unary action, binary comparison, or structural horn boundary.

- (3) **Opacity preservation.** The surface notion of sealing agrees with the raw opaque export. After sealing, later layers may use the normalized public interface, but not hidden transparent derivations except through explicitly exported fields or through the transparent operations admitted by counting normalization.
- (4) **Support preservation.** Normalization preserves the historical support of every primitive trace field. Repackaging may change notation, but it may not turn a field depending on layers  $i, j$  into one depending on a different public history.
- (5) **Primitive/derived preservation.** Presentation equivalence and normalization preserve the witnessed primitive/derived classification. A field is classified as derived only when the fixed derivability relation supplies a uniform construction from previously exported lower data.
- (6) **Cardinality preservation.** Payload size, primitive trace size, exported interface cardinality, and minimal opaque cost  $\mu$  are preserved by the raw-to-normalized signature bridge.
- (7) **Horn-computation preservation.** Higher structural obligations that the surface language treats as derived must elaborate to the structural horn-extension packages of Sections 4 and 5. In particular, their derived status must be witnessed by the cubical operations used in the computational replacement theorem, not merely asserted by a tag.
- (8) **Coupling preservation.** When a declaration advertises global action on an active interface footprint, the elaborated raw declaration must cover exactly that footprint. For the fully coupled recurrence theorem, this includes the per-site coverage assumptions of Section 2.

In this paper these clauses are proved or checked for the fixed raw extension calculus and for the small **SMod** surface instance below. For any other system, this list is the transfer target.

This formulation deliberately makes the broad transfer conjecture smaller and sharper. The paper proves a theorem inside  $C_{\text{ext}}$ ; the adequacy package records exactly what another system would have to preserve in order to inherit that theorem.

**3.2. Presentation steps and canonical normal forms.** The normalized public signature should not depend on how a programmer packages a public interface. A one-constructor record, a flattened telescope, and an equivalent iterated  $\Sigma$ -type should present the same sealed public data. Similarly, a derived alias should not become primitive merely because it was written explicitly. Presentation equivalence is therefore generated by a small list of local moves.

**Definition 3.3** (Presentation-step generators). Presentation equivalence for admissible declarations is generated by the following local moves on normalized public telescopes:

|                                     |                                                              |
|-------------------------------------|--------------------------------------------------------------|
| <code>reassocSigma</code>           | reassociate or flatten an iterated dependent sum;            |
| <code>splitRecord</code>            | split a single-constructor record or shell into fields;      |
| <code>bundleRecord</code>           | rebundle such a field telescope into a record or shell;      |
| <code>curryPi</code>                | turn a product argument into an iterated function telescope; |
| <code>uncurryPi</code>              | perform the inverse uncurrying step;                         |
| <code>transportField</code>         | move a field along an already exported equality;             |
| <code>transparentAliasInsert</code> | insert a transparently definable alias field;                |
| <code>transparentAliasDelete</code> | delete such an alias field;                                  |
| <code>duplicateDerivedDelete</code> | delete a duplicate derived trace field.                      |

Their reflexive, symmetric, and transitive closure is the relation

$$e \approx e'.$$

The list is also the scope of the raw bridge. It covers the raw constructors of Section 2: payload telescopes, action clauses, comparison and transport clauses, algebraic payload packages, export selections, and packaged horn-boundary clauses. It does not include arbitrary source-language rewrites or parser-level equivalence for Cubical Agda programs.

The point of this definition is conservative. Presentation equivalence forgets bureaucracy, not semantic obligations. A field may be removed only when it is an alias, a duplicate derived field, or a different presentation of the same public typing judgment.

**Proposition 3.4** (Counting normalization is stable under presentation steps). *Each generator of presentation equivalence preserves the canonical counting normal form up to telescope isomorphism, preserves primitive-versus-derived tags, and preserves historical support. Consequently, if  $e \approx e'$ , then*

$$N(e) \cong N(e'), \quad \text{Prim}(N_{\text{tr}}(e)) \cong \text{Prim}(N_{\text{tr}}(e')), \quad \mu(e) = \mu(e').$$

Reassociation, record splitting, rebundling, currying, and uncurrying change only the shape of the public telescope. After flattening the  $\Sigma$ -spine and normalizing the  $\Pi$ -telescope, both sides have the same atomic fields up to canonical renaming.

The transport move uses an equality already present in an earlier sealed public signature. It therefore changes the displayed type of a field but not the irreducible prior public data on which the field depends. Transparent alias insertion and deletion add or remove only fields whose inhabitants are uniformly synthesized from existing exported data. Such fields carry derived witnesses and do not contribute to primitive trace cost. Duplicate-derived deletion removes a second presentation of a field already classified as derived by such a witness.

Thus every generator preserves primitive classes and their historical support. Closure under reflexivity, symmetry, and transitivity gives the result for  $e \approx e'$ .

In the accompanying development this is the theorem-facing contract of `Metatheory/PresentationEquivalence.agda`, `Metatheory/TraceCostNormalForm.agda`, and `Metatheory/MuInvariance.agda`. The paper does not require a global confluence theorem for arbitrary source syntax. It requires uniqueness of the counted normal form for this generated presentation relation, which is exactly the invariant used by the later cardinality results.

**3.3. Elaboration, sealing, and public signatures.** We now connect admissible declarations to the candidate-and-obligation calculus of Section 2. The following theorem is the formal version of the pipeline displayed at the start of the section.

**Theorem 3.5** (Elaboration soundness for sealed exports). *Every admissible surface extension declaration  $e$  over a state  $B_n$  determines:*

(1) *an elaborated candidate*

$$X(e) := (X_{\text{core}}(e), G_{\text{obl}}(e));$$

(2) *a sealed layer  $L(e)$  exported through the opaque public interface selected by the export component  $\sigma$ ;*

(3) *a normalized public signature*

$$N(e) := N(\text{Elab}(e))$$

*obtained from that exported interface by counting normalization.*

*In particular, the candidate calculus of Section 2 is the image of admissible surface extensions under elaboration and sealing.*

Indeed, the payload telescope  $\Gamma_{\text{new}}$  elaborates to new core declarations, while  $\Gamma_{\text{act}}$  and  $\Gamma_{\text{cmp}}$  elaborate to the fields witnessing interaction with the active interface of  $B_n$ . The admissibility judgment says that these fields type-check and discharge the obligations of the declared coupling mode; the export selector  $\sigma$  determines the opaque public telescope; and counting normalization of that telescope gives  $N(e)$ . These are precisely the candidate and sealed layer used by Section 2.

**Theorem 3.6** (Adequacy of normalized public signatures). *Let  $e$  be an admissible surface extension over  $B_n$ , and let  $L(e)$  be its sealed export. Then:*

- (1) *every field of  $N(e)$  comes from a field of the sealed public interface of  $L(e)$ ; counting normalization introduces no new primitive public datum;*
- (2) *every field of  $N(e)$  is either a payload schema contributed by  $\Gamma_{\text{new}}$  or a trace schema arising from a resolved structural obligation of  $X(e)$ ;*
- (3) *every irreducible payload or trace schema visible after sealing occurs as a field of  $N(e)$ ;*
- (4) *the quantities*

$P(X(e)), \quad S(L(e)), \quad \Delta(X(e) \mid B_n), \quad \mu(e), \quad \text{ObSig}_k(X(e)), \quad \text{Ix}_k(X(e)), \quad \text{Real}_k(X(e))$   
*are exactly the corresponding payload family, public signature, abstract latency, minimal opaque cost, obligation signature, primitive index set, and realized obligation object computed from  $N(e)$  and its trace part.*

*Proof.* The first claim follows from the definition of sealing and counting normalization: the normalization procedure flattens the exported telescope, quotients presentation artifacts, and discards uniformly synthesized fields, but it does not invent primitive public data.

The second claim is the surface-language form of the payload/trace decomposition of Section 2. After elaboration and sealing, a visible field is either newly introduced mathematical content of the layer or public evidence that the new content interacts correctly with earlier opaque interfaces.

For the third claim, observe that counting normalization removes only presentation artifacts and transparently reconstructible fields. If an irreducible field is visible after sealing, it is not uniformly reconstructible from the remaining exported data by the admitted transparent operations. It therefore survives into  $N(e)$ .

The final claim is bookkeeping. Section 2 defines payload size, public schema set, integration latency, minimal opaque cost, obligation signatures, primitive index sets, and realized obligation objects from normalized public signatures and their induced trace obligations. Since  $N(e)$  is the canonical normalized signature of the sealed export, these abstract quantities agree with the corresponding surface quantities after elaboration.  $\square$

The theorem explains why the later sections can stop mentioning surface syntax. Once a declaration has crossed the adequacy bridge, all remaining claims are claims about normalized public signatures and their induced structural obligations.

**Irreducible public fields.** Let  $e$  be an admissible surface extension over  $B_n$ , and let  $\phi$  be an irreducible field of  $N(e)$ . If  $e' \approx e$ , then  $N(e')$  contains a canonically corresponding irreducible field  $\phi'$ . Equivalently, no presentation-equivalent declaration can delete an irreducible payload or primitive trace schema from the sealed public signature. This is immediate from the definition of presentation equivalence: it identifies elaborated sealed extensions only up to definitional equality, canonical telescope isomorphism, and transparent internal normalization. If  $\phi$  were absent from  $N(e')$ , then  $e'$  would either omit a public field present in  $e$ , contradicting  $e' \approx e$ , or reconstruct  $\phi$  uniformly from the remaining exported data, contradicting irreducibility.

**Canonical normalized presentations.** If  $e \approx e'$ , then  $N(e)$  and  $N(e')$  are canonically isomorphic, and their trace parts are canonically isomorphic. Consequently every presentation-equivalence class admits a canonical normalized public signature up to isomorphism, and

$$\mu(e) = |\text{Prim}(N_{\text{tr}}(e))|.$$

Here presentation equivalence means equality of public typing judgments up to the generated presentation steps, and counting normal form forgets exactly those differences. Hence  $N(e)$  and  $N(e')$  are canonically isomorphic, with corresponding trace subfamilies because the payload/trace split is part of the sealed public judgment. The preceding proposition prevents irreducible primitive trace from being deleted, so the minimum defining  $\mu$  is attained by the canonical normalized trace signature itself.

**3.4. Computational replacement on canonical trace presentations.** The previous results say that packaging changes do not alter  $\mu$ . The next theorem says something stronger and more important: an explicitly written higher structural field can vanish from primitive trace cost when a cubical term reconstructs it from its boundary. This is the formal expression of the phrase “derivedness is a theorem, not a tag.”

The theorem below does not itself construct the cubical term. That construction is supplied by the horn-to-open-box and open-box computation results of Sections 4 and 5. What the theorem proves is that, once the full replacement witness exists in the fixed derivability relation, the field is not allowed to remain in the  $\mu$ -minimal primitive public signature. This is the replacement rule later used by Definition 4.10.

**Theorem 3.7** (Computational replacement on canonical trace presentations). *Let  $X$  be a candidate over  $B_n$ , let  $k$  be a history depth, and let  $S$  be the canonical normalized trace signature attached to the depth- $k$  trace family of  $X$ . Let  $P$  and  $Q$  be two presented trace telescopes whose counting normal forms are  $S$ , and let  $\phi$  be a presented trace field of  $P$ .*

Suppose  $\phi$  is equipped with a replacement witness of the same shape as `DerivedTrace`: a lower-arity public boundary subfamily  $\partial\phi$ , a cubically synthesized term

$$t_\phi(\partial\phi)$$

built only from `hcomp`, `transp`, `hfill`, definitional equality, and the transparent structural operations already quotiented by counting normalization, together with typing and boundary equations, semantic preservation, allowed-dependency data, substitution stability, and public-normalization compatibility. Assume this witness shows that after passage to  $S$  the field corresponding to  $\phi$  is judgmentally reconstructed from lower-arity boundary data. Then:

- (1)  $P$  and  $Q$  are presentation-equivalent representations of the same canonical trace signature;
- (2) in the  $\mu$ -minimal part of the canonical normalized trace signature represented by  $S$ , the field corresponding to  $\phi$  is absent as an irreducible primitive trace field;
- (3) every other irreducible primitive trace field of  $P$  corresponds canonically to one of  $Q$ .

*Dependencies:* canonical normalized presentations, irreducible-field necessity, the fixed transparent derivability relation, and public-normalization compatibility of the replacement witness.

*Proof.* Both  $P$  and  $Q$  normalize to  $S$ , so they are two presentations of one canonical public trace interface. The hypothesis gives more than a label for  $\phi$ : it gives a full replacement witness. Its term  $t_\phi(\partial\phi)$  lies in the fixed transparent derivability relation, reconstructs the normalized image of  $\phi$  from lower-arity boundary data using only cubical Kan operations and presentation-transparent structural operations, and carries the typing, boundary, semantic-preservation, dependency, and substitution evidence needed for the canonical presentation relation.

Consequently the normalized image of  $\phi$  has a transparent derivedness witness rather than primitive status. Public-normalization compatibility identifies the presentation with explicit  $\phi$  and the presentation in which  $t_\phi(\partial\phi)$  is used instead. The  $\mu$ -minimal primitive signature retains only irreducible primitive trace fields, so  $\phi$  is excluded from that primitive part. Since primitive/derived status and support are preserved under passage between presentations of  $S$ , every other irreducible primitive field of  $P$  corresponds canonically to one of  $Q$ . Thus computational replacement changes the presentation of the trace telescope, not the primitive public content measured by  $\mu$ .  $\square$

This theorem is the bridge between the cubical argument and the accounting argument. The open-box theorem will show that the relevant higher structural fields have terms of the form  $t_\phi(\partial\phi)$ . The computational replacement theorem then turns that cubical fact into the public-signature fact that those fields do not contribute primitive trace cost.

**Remark 3.8** (Mechanized fixed raw bridge). The formal bridge in the accompanying Agda development is a three-part package for the fixed raw extension calculus.

- (1) Raw admissible declarations are connected, via elaboration and sealing, to canonical normalized signatures by `Metatheory/RawStructuralSyntax.agda`, `Metatheory/RawStructuralTyping.agda`, and `Metatheory/SurfaceNormalizationBridge.agda`.
- (2) Presentation-equivalent canonical trace normal forms preserve support, primitive cost, and  $\mu$  by the explicit presentation-step constructors in `Metatheory/PresentationEquivalence.agda` and `Metatheory/MuInvariance.agda`.

- (3) The resulting canonical normalized signatures are matched with the counted-interface quantities by the normalized-signature/counting bridge, and well-typed raw structural clauses are identified with the horn-generated trace image by `Metatheory/SurfaceToHornImage.agda`.

In particular, the computational replacement theorem is formalized at the level of canonical trace presentations. The bridge modules connect admissible raw declarations to that interface and to the counted abstract interface.

**3.5. The trace-cost dictionary.** The recurrence theorem later uses the symbols  $\mu_k$ ,  $\kappa_k$ ,  $\Delta$ , and  $S_{\text{bas}}(L_k)$ . Those symbols live at slightly different levels: surface declarations, normalized trace signatures, abstract obligation families, and primitive basis classes. The dictionary below records the only identifications used later.

Trace-cost dictionary. For every admissible surface extension declaration  $e$  over  $B_n$ ,

$$\mu(e) = |\text{Prim}(N_{\text{tr}}(e))| = \Delta(X(e) \mid B_n).$$

Adequacy of normalized public signatures says that the trace part of  $N(e)$  records exactly the irreducible public trace schemas arising from resolved obligations of  $X(e)$ , while canonical normalization identifies  $\mu(e)$  with the cardinality of the primitive part of that trace signature. The active-basis accounting of Section 2 reads those same primitive trace schemas as the abstract structural cost  $\Delta(X(e) \mid B_n)$ .

If  $e_k$  is the declaration whose sealing exports  $L_k$ , if  $\mu_k := \mu(e_k)$ , and if  $S_{\text{bas}}(L_k)$  is any basis family of  $L_k$ , then

$$|S_{\text{bas}}(L_k)| = \kappa_k + \mu_k.$$

For an admissible fully coupled declaration  $e_k$ , the quantities in

$$\mu(e_k) = |\text{Prim}(N_{\text{tr}}(e_k))| = \Delta(X(e_k) \mid B_{k-1}) = |\text{Str}(L_k)|.$$

are the same cardinality, read at their respective levels. Moreover, the primitive basis of the exported layer decomposes as

$$|S_{\text{bas}}(L_k)| = |S_{\text{core}}(L_k)| + |\text{Str}(L_k)| = \kappa_k + \mu_k.$$

Thus the recurrence later uses a cardinality identity obtained from normalized public signatures, not an additional semantic growth assumption.

**3.6. Factorization-complete public trace export.** The depth theorem is an arity theorem: it says that primitive structural trace stabilizes at binary comparison. The recurrence theorem needs one more ingredient. A binary comparison could still point far back in history, so arity depth two does not automatically imply a two-layer chronological window. The following definition isolates the extra export property needed to reuse recent public trace without reopening hidden derivations.

**Definition 3.9** (Factorization-complete public trace export). Let  $L_k$  be a sealed layer with  $k \geq 2$ . Its public trace export is *factorization-complete* when, after choosing any admissible declaration whose sealing exports  $L_k$  and reading its canonical normalized trace signature, the following data are available for the next remote-comparison step:

- (1) the adjacent unary bridge from  $L_k$  to  $L_{k-1}$ ;

- (2) the binary comparison trace witnessing the composite comparison from  $L_k$  to  $L_{k-2}$  through  $L_{k-1}$ ;
- (3) the degenerate or structural faces needed to assemble the corresponding open 3-box, generated transparently by definitional equality, transport along already exported equalities, and the structural operations already quotiented by counting normalization.

Equivalently, the next horn-extension problem involving  $L_k$  can be posed using only exported public trace fields and transparent structural operations, without reopening the hidden derivation that discharged the original sealing obligations.

**Remark 3.10** (Why the transparency boundary is not arbitrary). The phrase “generated transparently” is part of the theorem-facing boundary. It is not a license to declare inconvenient fields derived. A field is transparent only when the fixed derivability relation exhibits a uniform construction from already exported lower data, using definitional equality, canonical telescope isomorphisms, transport along exported equalities, or the cubical Kan operations admitted by the raw bridge.

Changing that derivability relation would change the measured primitive signature. This is why the paper treats the transparency boundary as part of adequacy rather than as informal notation. The two-layer window depends on a checkable claim: a remote comparison field must be computed by the horn term supplied later by the structural open-box theorem and the computational replacement theorem. If a proposed foundation cannot supply such a uniform term, the remote field remains primitive and the foundation does not instantiate the chronological factorization theorem.

**Theorem 3.11** (Sealed layers export factorization-complete trace data). *Every sealed layer  $L_k$  with  $k \geq 2$  in the fixed fully coupled cubical extension calculus has factorization-complete public trace export.*

Dependencies: payload and trace decomposition, normalized public signature adequacy, irreducible-field necessity, and the transparency boundary for generated structural faces.

*Proof.* Choose an admissible declaration  $e_k$  whose sealing exports  $L_k$ . By the payload/trace decomposition, the public export of  $L_k$  consists of new payload together with trace. By adequacy of normalized public signatures, every irreducible trace schema visible after sealing appears in the canonical normalized trace signature  $N_{\text{tr}}(e_k)$ .

Because  $L_k$  is sealed as a globally acting layer against the active interface, its resolved trace obligations include the adjacent action on  $L_{k-1}$  and the binary comparison against the already exported comparison between  $L_{k-1}$  and  $L_{k-2}$ . Hence the corresponding unary bridge and binary comparison trace occur as public fields of the normalized trace signature.

The remaining faces needed for the next open 3-box are not extra primitive obligations. They are degenerate composites, reindexings, and transports along already exported equalities, so they are generated by the transparent structural operations admitted in counting normalization. Presentation equivalence cannot delete the irreducible unary or binary trace fields just identified; it can only rewrite them into canonically isomorphic normal form. Therefore the factorization data are exported stably at the level of canonical normalized public signatures and can be reused without reopening the hidden sealing derivation.  $\square$

**Remark 3.12** (Use of the bridge in later sections). Sections 4 and 5 invoke surface syntax only through the adequacy bridge just described. Once an admissible declaration has been identified with its canonical normalized signature and the corresponding counted-interface



data, the later horn, depth, factorization, and recurrence theorems are applications of the abstract candidate-and-obligation calculus.

#### 4. NORMAL FORMS FOR STRUCTURAL TRACE

The previous section ended at the point where every admissible declaration has been replaced by a canonical normalized public signature. We now forget the surface presentation and look only at the structural trace fields that survive this passage to normal form. This section is the bridge between the accounting calculus of Sections 2–3 and the cubical depth theorem of Section 5.

The bridge has one central purpose. It explains why the later phrase “higher trace is derived” is not merely a syntactic tag. A normalized trace field is not classified by its textual shape alone. It is classified by the semantic role it plays in the sealed interface, by the historical support on which its typing derivation genuinely depends, and by whether there is a uniform construction of the field from already exported lower data.

After normalization, every structural trace field generated by the sealing discipline has one of three roles:

unary action trace,      binary comparison trace,      higher structural horn package.

The first two roles may contribute irreducible primitive public trace. The third role may remain present in the exact realized obligation object, but in the cubical calculus it is carried by an explicit horn-computation witness and therefore contributes no primitive field to a  $\mu$ -minimal public signature.

This is the section-level slogan:

Normal form does not erase higher structural obligations; it exposes them as  
horn packages whose derivedness must be witnessed.

**4.1. The three trace roles.** Let  $L_{n+1}$  be a sealed layer over a normalized prior state  $B_n$ , and let  $N_{\text{tr}}(L_{n+1})$  be the trace part of its canonical normalized public signature. A field of  $N_{\text{tr}}(L_{n+1})$  is structural only when its job is to integrate the new payload with previously exported opaque interface data. The same mathematical expression may have a different status when it is authored as new payload. For example, a path constructor contributed by a new higher inductive type is payload; a path witnessing that a new globally acting operation respects an old exported equivalence is structural trace.

**Definition 4.1** (Normal structural trace role). A normalized structural trace field  $t \in N_{\text{tr}}(L_{n+1})$  has one of the following roles.

- (1) **Unary action trace.** The field records how the new payload acts on one old opaque public site. Its historical support contains one prior public interface component after counting normalization.
- (2) **Binary comparison trace.** The field records compatibility between two public routes already determined by unary action, transport, old comparison data, or adjacent composites through those data. Its historical support contains two independent public interface components, or an adjacent binary composite whose two components remain visible in normal form.

| Normal role         | Semantic content                                                                                          | Historical support                                                                              | Primitive status                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Unary action        | Action of the new payload on one exported interface site.                                                 | One prior public field or one primitive class of an old opaque signature.                       | Primitive candidate unless transparently generated.                                                    |
| Binary comparison   | Compatibility between two already public routes: actions, transports, equivalences, comparison witnesses. | Two prior public fields, or an adjacent composite trace through them.                           | Primitive candidate; necessary in general by the univalent swap obstruction.                           |
| Higher horn package | Boundary, missing face, and filler data for a higher structural obligation generated by sealing.          | A horn boundary assembled from unary and binary public trace plus transparent structural faces. | Derived in the fixed cubical calculus when filled by <b>hcomp</b> , <b>transp</b> , and <b>hfill</b> . |

Table 1: Normal roles for structural trace fields.

- (3) **Higher structural horn package.** The field records the data of a marginal higher coherence problem generated by sealing. Its visible boundary is assembled from already exported unary and binary trace, together with degenerate faces, reindexings, and transports that counting normalization treats as transparent. Its missing face and filler are represented as an open-box extension package.

The roles should not be read as a hierarchy of textual arities. A field whose type begins with a long telescope may still be unary if that telescope ranges over parameters inside one old opaque interface. Conversely, a short-looking comparison may be binary if its typing derivation uses two independent old public exports. The invariant is historical support after all prior layers have been collapsed to their opaque normalized signatures.

The table is an accounting table, not a claim that only one-dimensional and two-dimensional mathematics exists. Higher-dimensional payload may be primitive payload. What is at issue is whether a higher field is forced only to make a sealed structural action coherent with old opaque public data. Such fields are the horn packages analyzed below and in Section 5.

**4.2. Unary action trace.** Unary action trace is the most direct form of public integration debt. A globally acting sealed layer advertises that its payload can be applied to the active interface already exported by the foundation. For each active basis site that lies in the declared footprint, the normalized public signature must contain either a primitive action field or a transparent construction of that action from previously exported data.

**Definition 4.2** (Unary action normal form). A structural trace field  $a$  is in unary action normal form when there are an old primitive basis site  $s \in S_{\text{bas}}(L_i)$  and a new payload component  $X$  such that, after canonical normalization,  $a$  has the form

$$a_s : \text{Act}_X(s),$$

up to the telescope isomorphisms and transparent transports quotiented by counting normal form. Its support is  $\text{Supp}(a_s) = \{L_i\}$ .

The notation  $\mathbf{Act}_X(s)$  is schematic. In a universe example it may be an action on codes; in a modality example it may be a decoding law; in an algebraic interface it may be the image of a generator. What matters is that only one old opaque public site is being acted on.

Unary action trace is usually primitive because later layers cannot reconstruct objectwise action on an opaque site unless that action has been exported. A field of unary shape is nevertheless not counted as primitive if it is transparently generated. For instance, an action on a record projection may be definitionally induced by the action on the record itself, and an action on a rebundled presentation may be a canonical telescope isomorphism of an already exported action. The primitive part of the unary trace consists only of the irreducible classes left after these transparent identifications.

**Lemma 4.3** (Unary support stability). *If two admissible presentations of a sealed layer are presentation-equivalent, then the unary action fields of their normalized public trace signatures correspond bijectively up to primitive-class equivalence. In particular, presentation equivalence cannot turn an irreducible unary action on one old opaque basis site into a trace field supported on a different old layer.*

The normalization moves of Section 3—telescope rebracketing, record–telescope conversion, definitional expansion and contraction, transparent transport along exported equalities, and deletion of uniformly synthesized fields—do not change which old opaque primitive class is being acted on. Generated or rebundled actions may leave the primitive count, but an irreducible action on a distinct old basis site cannot move to another site unless the normalized old interface already identifies them, exactly as recorded by the primitive-class quotient.

This lemma is one of the reasons the counting invariant  $\mu$  is computed only after normalization. Before normalization, a programmer may write several aliases for the same action or hide an action inside a record field. After normalization, the unary action role is read from the old opaque support and is independent of those packaging choices.

**4.3. Binary comparison trace.** Binary comparison trace is the first genuinely cubical obstruction. It records that two public routes agree. The simplest route comparison has the shape

$$r_0, r_1 : P \quad \text{and} \quad c : r_0 = r_1,$$

but the actual endpoints  $r_0$  and  $r_1$  may themselves be composites of action fields, old transports, decoding equations, univalence paths, or adjacent comparison trace.

**Definition 4.4** (Binary comparison normal form). A structural trace field  $c$  is in binary comparison normal form when, after canonical normalization, it compares two public composites whose irreducible support uses exactly two old public components. Schematically,

$$c_{s,t} : \mathbf{Cmp}_X(s, t; r_0, r_1),$$

where  $s$  and  $t$  are old opaque public sites,  $r_0$  and  $r_1$  are the two normalized routes built from unary action and previously exported trace, and  $\text{Supp}(c_{s,t}) = \{s, t\}$  after quotienting transparent aliases.

Binary comparison trace is not redundant in univalent cubical settings. Objectwise action on the endpoints of an old equivalence does not by itself prove naturality along that equivalence. The two-point swap obstruction of Section 5 makes this precise: the old interface may contain a public path generated by a nontrivial equivalence, and a sealed operation must say how its action respects that path. Therefore binary comparison fields remain primitive candidates in the normal form.

**Remark 4.5** (Why binary comparison is not merely two unary actions). A unary field tells us how  $X$  acts on  $s$ , and another unary field tells us how  $X$  acts on  $t$ . A binary field tells us that these actions commute with a public comparison between  $s$  and  $t$ . In a univalent foundation the comparison may carry real information: an equivalence can be transported into an equality of types, and the action of  $X$  must be compatible with that equality. Thus the binary role records a naturality square, not a pair of isolated objectwise values.

Binary fields are nevertheless subject to the same normalization discipline as unary fields. A comparison obtained by whiskering an already exported comparison with definitional maps is derived. A comparison whose two routes become judgmentally equal after normalization is derived. A comparison duplicated by a rebundled record field is not counted twice. But if the old public interface contains an irreducible comparison and the new sealed action must respect it, the resulting naturality witness is primitive unless the derivability relation exhibits a uniform construction.

**Lemma 4.6** (Binary comparison support stability). *Presentation equivalence preserves the primitive-class support of irreducible binary comparison trace. It may change the displayed routes by canonical isomorphism or transport, but it cannot eliminate a genuine binary comparison unless the comparison is transparently constructed from already exported lower trace.*

The same normal-form argument applies to binary support. Its two endpoints are route expressions in the normalized public interface; presentation moves may reassociate, unfold, or transport them, but after quotienting by transparent operations the pair of independent old supports remains visible. If the comparison disappears, the presentation relation has supplied a transparent construction from previously exported data, so the field is derived and contributes no primitive trace. Otherwise it remains an irreducible binary class.

**4.4. Higher trace as a structural horn package.** The third normal role is the one used by the upper-bound proof. A higher structural field is not represented as an arbitrary choice of a filler for a completely fixed closed boundary. It is represented as a marginal extension problem. The lower faces are already public. One remote comparison face is not yet public. The obligation is to produce that missing face and a filler making the resulting open box commute.

This distinction is crucial in a cubical setting. The space of fillers of a closed higher boundary need not be contractible. Cubical types may have genuine higher homotopy. The paper therefore does not claim that all higher fillers are unique. It claims that the sealing-generated higher structural obligation has open-box extension form, and that the fixed cubical operations compute the missing structural data needed for public normalization.

**Definition 4.7** (Structural horn grammar in normal form). The higher structural fields generated by sealing are the derivations of a judgment

$$\Gamma \vdash H : \text{SHorn}_k(X) \quad (k \geq 3),$$

where  $\Gamma$  is a normalized public context for the already sealed layers and the lower-arity trace of  $X$ . Such a derivation carries the following projections:

$$\begin{aligned} \partial(H) &: \text{Real}_{k-1}(X), \\ \text{supp}(H) &\subseteq \{X, L_n, \dots, L_{n-k+1}\}, \\ \text{miss}(H) &: \text{Type}, \\ \text{fill}(H) &: \text{miss}(H) \rightarrow \text{Type}, \\ \text{export}(H) &: \{\text{public}, \text{transparent}\}. \end{aligned}$$

It also carries a substitution action: for every cubical substitution  $\sigma : \Delta \rightarrow \Gamma$ , there is a reindexed derivation

$$\Delta \vdash H[\sigma] : \text{SHorn}_k(X[\sigma])$$

whose projections are the corresponding reindexings of  $\partial(H)$ ,  $\text{miss}(H)$ ,  $\text{fill}(H)$ , support, and export status.

The constructors are the following.

| Constructor       | Boundary data                                                                                                         | Missing face                                                                                                                        | Filler and public status                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Remote comparison | A boundary $\partial : \text{Real}_{k-1}(X)$ assembled from public trace, public transparent transports/degeneracies. | The direct comparison face between $X$ and the newly posed remote layer $L_{n-k+1}$ , compatible with the endpoints of $\partial$ . | A $k$ -cell filling the open comparison box; public when the surface presentation exports it, transparent after replacement. |
| Degenerate horn   | A lower structural horn with one interval direction degenerated, together with the degeneracy map.                    | The degenerate remote face obtained by applying the degeneracy to the lower missing face.                                           | The degenerate filler; transparent unless explicitly exported by the surface syntax.                                         |
| Transported horn  | A structural horn $H$ and a public path or equivalence used to transport one boundary face.                           | The transported missing face, with endpoint compatibility transported along the same public path.                                   | The transported filler; export status inherited from $H$ .                                                                   |
| Reindexed horn    | A structural horn $H$ and a substitution $\sigma$ of normalized public contexts.                                      | $\text{miss}(H)[\sigma]$ .                                                                                                          | $\text{fill}(H)[\sigma]$ ; export status inherited from $H$ .                                                                |

The element presented by such a derivation is the telescope

$$\sum_{\gamma : \text{miss}(H)} \text{fill}(H)(\gamma).$$

The remote-comparison constructor is the only constructor that introduces a genuinely new higher structural obligation. The degenerate, transported, and reindexed constructors record the closure conditions needed for substitution stability and presentation invariance. They do not introduce new payload and do not make derivedness true by declaration.

**Proposition 4.8** (Structural horn induction). *To prove a property  $P(H)$  of every sealing-generated structural horn, it is enough to prove the remote-comparison case and show that  $P$  is preserved by the degenerate, transported, and reindexed constructors. If  $P$  includes*

*equations for  $\partial(H)$ ,  $\text{miss}(H)$ ,  $\text{fill}(H)$ , *support*, or *export status*, the preservation clauses must use the displayed substitution and transport actions.*

This is the ordinary eliminator for the grammar of Definition 4.7, with the projection equations included as computation rules for the four constructors.

The notation is intentionally close to the later definition of  $H_k(\partial)$ . This section classifies the normal role; Section 5 proves by the induction principle that the corresponding cubical open-box extension object is contractible and substitution-stable in the fixed calculus.

**Remark 4.9** (The boundary is public). The boundary  $\partial_h$  is not hidden derivation state. It is built from the normalized public signature exported by earlier layers and from the lower-arity trace of the current sealed layer. This is why a later layer can reuse the horn computation without reopening the opaque proof that produced the current layer. If the boundary of a proposed higher field depends on private construction details not present in the normalized public interface, then that field is not a well-formed structural horn package for the present calculus.

A useful mental picture is the first nontrivial case. Suppose the new layer has unary action on a recent public site and binary comparison against an adjacent public comparison. When a further remote layer is exposed, the already exported unary and binary data determine all but one face of a three-dimensional compatibility box. The remaining face is the direct remote comparison. The structural horn package asks for that face together with a filler. The cubical theorem later says that this package is computed from the displayed boundary by the ambient Kan operations.

**4.5. Derivedness witnesses.** The word “derived” is safe only when accompanied by a witness. In this paper a higher structural field is derived because there is a uniform computation that reconstructs it from its lower boundary. The witness is part of the normalized role, and the proof of its adequacy is discharged by the open-box theorem of Section 5.

**Definition 4.10** (Derived trace witness). Let  $h$  be a structural horn package with visible boundary  $\partial_h$ . A *derived trace witness*, written  $\text{DerivedTrace}(h)$ , consists of:

- (1) lower public boundary data  $\partial_h$  assembled from already exported unary and binary trace plus transparent structural operations;
- (2) a replacement cubical term

$$t_h(\partial_h) : \sum_{\gamma : \Gamma_{\text{miss}}(\partial_h)} \text{Fill}_{\partial_h}(\gamma),$$

constructed uniformly from the visible boundary using the structural operations admitted by the fixed calculus, in particular composition, transport, and filling;

- (3) typing and boundary agreement equations showing that the computed missing face and filler have the presented field’s type and restrict to the displayed visible faces;
- (4) a semantic preservation proof identifying the explicit presentation of  $h$  with the replacement term in the realized obligation object;
- (5) an allowed-dependency proof that  $t_h$  uses only lower boundary data and the fixed transparent cubical operations;
- (6) substitution stability: for every substitution  $\sigma$ , the reindexed computation  $\sigma^*t_h(\partial_h)$  agrees with the computation on the reindexed boundary  $t_h(\sigma^*\partial_h)$ , up to the canonical equality of the cubical structure;

- (7) public-normalization compatibility: replacing an explicitly presented field  $h$  by  $t_h(\partial_h)$  gives a presentation-equivalent normalized public signature in which  $h$  is not a primitive trace field.

The term  $t_h$  is the formal content behind the phrase “computed horn”. In the fixed cubical calculus, Section 5 constructs  $t_h$  by turning the structural horn package into an open-box extension problem and applying the cubical composition and filling operations. The transport component aligns the computed face with the already exported comparison data. The filling component supplies the interior cell.

**Remark 4.11** (Derived is not a color on syntax). It would be unsatisfactory to define a raw syntax constructor `derivedHorn` and then conclude that all higher trace is derived. The normal-form judgment used here is stronger. A higher field may be assigned the horn role only when its boundary is public and its derivedness obligation is a theorem-facing computation problem. For the fixed cubical calculus, the computation is discharged by the open-box theorem. For a different surface language or foundation, the adequacy package must preserve this witness.

This is also where the paper’s two notions of depth begin to separate. A horn package may remain present as an exact obligation factor: the realized object still contains the type of missing-face-and-filler data. Once a derived trace witness has been supplied, however, that package is absent from the primitive part of the normalized public signature. Thus higher exact factors can be present and contractible while higher primitive public fields are eliminated.

Primitive status of the three roles. Assume every normalized higher structural horn package in a sealed declaration carries a derived trace witness. Then the primitive part of the normalized structural trace signature is contained in the union of unary action trace and binary comparison trace:

$$\text{Prim}(N_{\text{tr}}(L_{n+1})) \subseteq \text{Act}_{\text{prim}}(L_{n+1}) \sqcup \text{Cmp}_{\text{prim}}(L_{n+1}).$$

Equivalently, no higher structural horn package contributes an irreducible primitive public trace field to  $\mu$ . By the trichotomy of normal roles, each normalized structural trace field is unary, binary, or horn-shaped. Unary and binary fields remain primitive exactly when they are not removed by transparent generation, canonical telescope isomorphism, or presentation equivalence. A horn-shaped field with a derived trace witness is uniformly replaceable by  $t_h(\partial_h)$ , and public-normalization compatibility identifies the presentation with the explicit field with the one in which the field is synthesized. Counting normal form therefore assigns no primitive trace cardinality to that horn field.

The proposition is deliberately conditional. Section 4 supplies the normal-form classification and states what a derivedness witness must contain. Section 5 proves that the structural horn packages generated by the fixed cubical sealed-extension calculus have such witnesses.

**4.6. Payload is not reclassified as trace.** The theorem is about public structural integration obligations, not about a collapse of higher mathematics. This distinction is easiest to lose precisely at the point where higher horn packages appear. The normal-form discipline therefore includes a payload exclusion rule.

**Definition 4.12** (Payload exclusion). A field belongs to payload rather than structural trace when it is authored as new semantic content of the sealed layer and is not forced merely by the requirement that the new payload act coherently on the old opaque interface. In particular, the following are payload when introduced as part of the new layer:

- (1) new operations, type formers, modalities, or universe constructors;
- (2) algebraic laws internal to the new operation;
- (3) higher inductive constructors or higher path constructors contributed as new mathematical structure;
- (4) user-specified higher cells whose boundary is internal to the new payload rather than assembled from prior public trace.

Such fields are counted by the payload invariant  $\kappa$ , not by the structural trace cost  $\mu$ , unless the sealing discipline separately requires public integration trace for their action on old interfaces.

The exclusion rule prevents a misleading reading of the depth-two theorem. A layer may export a primitive three-dimensional constructor as payload. The theorem says only that the additional structural coherence required to integrate that constructor with the existing sealed interface is normalized as unary action, binary comparison, and derived horn computation. It does not say that the constructor itself is derived or absent.

No payload-to-trace leakage. Canonical normalization preserves the distinction between payload fields and structural trace fields. A payload field can be accompanied by structural trace explaining its action on old interfaces, but the payload field itself is not moved into  $N_{\text{tr}}$  merely because it has higher arity or higher dimension. The normalized public signature decomposes into payload and trace parts before primitive trace cost is computed. Presentation moves may rebundle a payload field, unfold a record containing it, or transport it along an exported definitional equality, but they do not change its semantic role as new content of the sealed layer. A field enters the trace part only when its typing judgment is an integration obligation against previously exported opaque data; higher-dimensional payload may increase  $\kappa$ , while induced compatibility fields are classified separately.

**4.7. Role uniqueness and presentation invariance.** The normal roles are useful only if they are stable under the presentation moves already quotiented by the adequacy bridge. The following proposition records the invariant that is used implicitly whenever later sections reason about “the” action part, comparison part, or horn part of a normalized trace signature.

**Proposition 4.13** (Trace-role normal form). *Every structural field in the canonical normalized trace signature of an admissible sealed layer has a unique normal role, up to the primitive-class quotient of Section 2 and the presentation equivalence of Section 3. Presentation-equivalent declarations have canonically isomorphic role decompositions:*

$$N_{\text{tr}}(e) \cong N_{\text{tr}}(e') \implies \begin{cases} \text{Act}(e) \cong \text{Act}(e'), \\ \text{Cmp}(e) \cong \text{Cmp}(e'), \\ \text{Horn}(e) \cong \text{Horn}(e'). \end{cases}$$

*The isomorphisms preserve historical support and primitive/derived status.*



Canonical normalization flattens the exported public telescope, quotients by transparent presentation moves, and records primitive classes. In this normal form, the irreducible support and shape of a structural typing derivation determine the role: support of size one is unary action, support of size two with route-comparison shape is binary comparison, and the inductive sealing grammar gives horn packages when a higher coherence problem is presented as a public boundary plus missing face and filler. Presentation equivalence preserves these data, and the three cases are disjoint after opacity, so the role decomposition is unique and presentation invariant.

**Remark 4.14** (Adjacent composites). The phrase “support of size two” should be read after normalization, not by counting the number of names in a displayed term. An adjacent composite may mention several textual components, but if all but two are transparent degeneracies or transports generated from the normalized public signature, the field is binary. Conversely, a term with few variables may still be higher if its typing derivation depends irreducibly on a horn boundary assembled from several independent prior exports.

## 5. OBLIGATION DEPTH, OPEN BOXES, AND EXACT DEPTH TWO

The preceding sections put the public trace of a sealed extension into normal form. We now ask the main semantic question: once that normal form has been reached, how far up the coherence tower must the exact sealing obligation continue to grow?

There are two tempting but wrong answers. The first says that unary action should be enough. This is false in a univalent setting, because an extension must respect exported paths and equivalences, not merely exported objects. The boolean swap obstruction below is an explicit witness. The second says that if binary comparison is needed, then an infinite tower of higher primitive coherence fields should also be needed. This is also false for the fixed cubical sealed-extension calculus. The higher structural obligations generated by sealing are not arbitrary closed-boundary filler spaces. They are open-box extension problems whose missing face is part of the extension data. Cubical composition and filling supply a canonical extension.

Thus Section 5 proves a dichotomy:

Unary trace is too weak, binary trace is necessary, and every sealing-generated structural obligation above binary trace is an open-box extension factor computed by the cubical core.

The section is deliberately careful about levels. Exact realized obligation objects may still contain higher factors. Those factors are contractible. Minimal public signatures, which count only irreducible primitive trace, do not contain those factors as primitive fields. Chronological locality is a third issue again: a binary field could in principle point far back into history. A separate factorization theorem below turns the arity-depth result into the two-layer window used in the counting theorem.

**5.1. Exact obligations, primitive signatures, and historical arity.** Let a candidate extension be sealed against a normalized historical interface. For each allowed depth bound we need both a syntactic signature of public obligations and the actual semantic object whose inhabitants discharge those obligations. The two should not be confused. Syntax determines what may be presented and counted; the realized object determines whether the exact obligation space has stabilized.

**Definition 5.1** (Induced obligation signatures and realized objects). Fix a candidate  $X$  sealed against a normalized historical interface  $\Gamma$ . For a depth parameter  $k$ , the normalized obligation signature up to historical arity  $k$  is a finite dependency-ordered telescope

$$\text{ObSig}_k(X) = (i_1 : A_{i_1}, \dots, i_m : A_{i_m}(i_1, \dots, i_{m-1})),$$

whose entries are exactly the normalized public obligations of arity at most  $k$ . Each entry carries its trace role, support, export status, and, when it is derived, a derivation witness rather than an unverified tag. The realized obligation object is the interpretation of this telescope in the cubical semantics:

$$\text{Real}_k(X) := \sum_{x_{i_1} : A_{i_1}} \sum_{x_{i_2} : A_{i_2}(x_{i_1})} \cdots \sum_{x_{i_m} : A_{i_m}(x_{i_1}, \dots, x_{i_{m-1}})} \mathbf{1}.$$

Equivalently,  $\text{Real}_k(X)$  is the dependent record of choices discharging all obligations in  $\text{ObSig}_k(X)$ . We also write:

$$\text{Ix}_k(X), \quad \text{Prim}_k(X), \quad \text{Der}_k(X).$$

These name the primitive index set, primitive irreducible sub-signature, and derivation-witnessed sub-signature associated to  $\text{ObSig}_k(X)$ :

|                     |                                                                                 |
|---------------------|---------------------------------------------------------------------------------|
| $\text{ObSig}_k(X)$ | the finite normalized obligation signature visible up to historical arity $k$ ; |
| $\text{Ix}_k(X)$    | the finite set of primitive schema indices in $\text{ObSig}_k(X)$ ;             |
| $\text{Real}_k(X)$  | the dependent record interpreting $\text{ObSig}_k(X)$ ;                         |
| $\text{Prim}_k(X)$  | the primitive irreducible sub-signature;                                        |
| $\text{Der}_k(X)$   | the sub-signature equipped with explicit derived-trace witnesses.               |

The signature  $\text{ObSig}_k(X)$  is read in the counting normal form of Section 2: canonical telescope isomorphisms are quotiented, transparent reindexings are not counted as new fields, and primitive status is assigned only after the derived-witness construction has failed or is unavailable in the fixed normalization relation. The index set  $\text{Ix}_k(X)$  indexes only primitive fields. The object  $\text{Real}_k(X)$ , by contrast, is the exact semantic obligation object.

The slogan is:

$$\begin{aligned} \text{Real}_k(X) &\text{ measures exact semantic obligation,} \\ \text{Prim}_k(X) &\text{ measures primitive public syntax.} \end{aligned}$$

The main stabilization theorem is a theorem about  $\text{Real}_k(X)$ . The trace-cost bound is a theorem about  $\text{Prim}_k(X)$  and about the trace part of normalized public signatures. These theorems interact, but neither one is merely a restatement of the other.

When a higher structural problem appears with one face still free, its exact semantic object is represented by an extension type for the missing face together with a filler. It is not represented by the type of fillers for a completely fixed closed boundary. This convention is what makes the later open-box theorem applicable.

Because interfaces grow cumulatively, the signatures form an inclusion system. If  $j \leq k$ , forgetting all obligations of arity greater than  $j$  induces a restriction map

$$\text{res}_{j,k}(X) : \text{Real}_k(X) \longrightarrow \text{Real}_j(X).$$

These maps are telescope projections, so they satisfy

$$\text{res}_{k,k} = \text{id}, \quad \text{res}_{i,j} \circ \text{res}_{j,k} = \text{res}_{i,k} \quad (i \leq j \leq k).$$

The corresponding primitive index sets have inclusions

$$\text{Ix}_1(X) \hookrightarrow \text{Ix}_2(X) \hookrightarrow \text{Ix}_3(X) \hookrightarrow \dots,$$

before derived replacement is applied. The depth question asks when adding higher-arity entries changes no exact obligation object, up to the canonical equivalences constructed below.

Throughout this section, a *canonical equivalence*

$$\text{Real}_k(X) \simeq \text{Real}_j(X)$$

means an equivalence of the dependent telescope objects just defined. One map is always the restriction projection  $\text{res}_{j,k}(X)$ . The inverse is constructed by adjoining, in dependency order, the canonical centers of the higher structural extension fibers. The two round trips are the telescope equalities or homotopies obtained from the contraction of those fibers. Thus the stabilization theorem is not a cardinality statement and not a presentation-equivalence quotient; it is an equivalence of cubical dependent records, natural in the substitutions for which the structural open-box centers are stable.

**Remark 5.2** (Five statuses for a higher-looking field). The paper uses five different statuses that should not be collapsed.

| Status                                     | Meaning                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Present obligation type                    | A normalized obligation type genuinely occurs in some $\text{Real}_k(X)$ .                                                                                         |
| Contractible obligation type               | The type occurs in $\text{Real}_k(X)$ , but its fiber is canonically contractible.                                                                                 |
| Transparently derived field                | A field is part of a typing judgment or presentation, but is uniformly reconstructed from exported lower data by cubical computation, transport, or normalization. |
| Eliminated from a $\mu$ -minimal signature | The field may appear in a presentation, but not as a primitive field in the canonical normalized public signature measured by $\mu$ .                              |
| Absent from the exact obligation object    | No normalized obligation type of that kind occurs in the relevant $\text{Real}_k(X)$ .                                                                             |

The higher horn factors considered below are present in exact obligation objects, are contractible as open-box extension fibers, are transparently derived, and are eliminated from  $\mu$ -minimal public signatures. They are not absent from the exact semantics.

**Definition 5.3** (Historical support and arity). Let  $o$  be a normalized structural obligation. Its support, written  $\text{Supp}(o)$ , is the set of sealed layers whose atomic exported schemas occur irreducibly in the counting normal form of the typing derivation of  $o$ , after every prior layer has been collapsed to its opaque normalized interface. The historical arity of  $o$  is

$$a(o) := |\text{Supp}(o)|.$$

Historical arity counts independent prior interfaces, not textual size. A long formula that mentions many parameters inside one opaque old signature can still be unary. Conversely, a short comparison can be binary if its typing derivation depends irreducibly on two independent sealed exports.

**Lemma 5.4** (Arity-to-cell correspondence). *In the structural-integration grammar of the fixed calculus, an irreducible obligation of historical arity  $k$  has the shape of a  $k$ -dimensional structural coherence cell. Arity one gives action or transport data. Arity two gives a naturality square or comparison homotopy. Arity  $k \geq 3$  gives a marginal higher horn-extension cell.*

*Proof.* Unary action clauses mention one prior opaque signature, so their normalized obligations are single action, substitution, or transport clauses. Binary comparison clauses mention two independent prior signatures and compare the two unary composites through them, giving a square or homotopy. The higher horn clauses are generated inductively: passing from arity  $k - 1$  to arity  $k$  exposes one further remote layer and asks for the missing comparison face plus a filler of the resulting open box. Since support is computed after opacity and counting normalization, transparent aliases, record rebundlings, and transports along already exported equalities cannot increase this arity.  $\square$

**Definition 5.5** (Obligation depth and minimal-signature depth). A fixed foundation or extension calculus has *obligation depth*  $d_{\text{obl}}$  if  $d_{\text{obl}}$  is the least integer such that, for every admissible candidate  $X$  and all  $k \geq d_{\text{obl}}$ , there is a canonical equivalence

$$\text{Real}_k(X) \simeq \text{Real}_{d_{\text{obl}}}(X).$$

It has *minimal-signature depth*  $d_\mu$  if  $d_\mu$  is the least integer such that every admissible surface declaration  $e$  is presentation-equivalent to a declaration  $e'$  whose primitive trace sub-signature contains no irreducible primitive historical field of arity greater than  $d_\mu$ .

When this paper says simply “coherence depth,” it means obligation depth unless explicitly qualified. The minimal-signature depth is a public-accounting invariant. It says which fields remain primitive after normalization; it does not by itself say that exact higher obligation objects are absent.

**Definition 5.6** (Chronological window). A fixed extension calculus has chronological window size at most  $d$  if every admissible sealing step at stage  $n + 1$  has the following property after passage to the canonical normalized public signature: every surviving primitive trace field has support contained in

$$\{L_n, L_{n-1}, \dots, L_{n-d+1}\},$$

and every comparison with a remote layer  $L_{n-r}$ , for  $r \geq d$ , is presentation-equivalent to a trace term factored through the exported normalized signatures of those  $d$  recent layers.

A bound on arity is not yet a chronological bound. An arity-two comparison could compare a new layer directly with a very old one. The recent-history theorem below supplies the extra factorization needed before the recurrence theorem in Section 6 can count only a two-layer window.

**Remark 5.7** (Theorem dependency diagram). The main proof chain separates exact semantic obligations, public-signature minimality, chronological support, and later counting arithmetic:

$$\begin{aligned} \text{fixed grammar} &\Rightarrow \text{horn induction and horn-to-open-box adequacy} \\ &\Rightarrow \text{total OpenExt contractibility} \\ &\Rightarrow \text{exact stabilization of } \text{Real}_k \\ \text{total OpenExt} &\Rightarrow \text{DerivedTrace witnesses} \\ &\Rightarrow \text{replacement in normalized signatures} \\ &\Rightarrow d_\mu \leq 2 \\ \text{swap obstruction} &\Rightarrow d_{\text{obl}} \geq 2 \\ \text{recent-history factorization} &\Rightarrow \text{two-layer counting window.} \end{aligned}$$

The recurrence theorem in Section 6 starts only after the last implication and after a separate full-coupling hypothesis. It is not an input to the cubical depth theorem.

**5.2. The cubical upper bound: higher structural obligations are open boxes.** We now specialize to the fixed sealed-extension calculus over the chosen cubical core. The ambient core has interval structure, transport, partial elements, homogeneous and dependent composition, filling, and univalence. Full coupling is not used in this subsection. The upper bound depends on the structural horn grammar and on cubical open-box computation; full coupling returns only in Section 6, when the two-layer window is counted.

The important semantic clarification is this: the paper does not claim that arbitrary higher filler spaces in a univalent type theory are contractible. They are not. The relevant object is more specific. It is the theorem-facing extension type of a structural open box whose visible faces are fixed by prior public trace and whose missing remote face is still part of the extension data.

**Definition 5.8** (Structural horn-extension package). Let a candidate  $X$  be sealed at stage  $n + 1$ , and let  $k \geq 3$ . The remote layer exposed by the depth- $k$  step is

$$R_k(n) := L_{n-k+1}.$$

For a boundary  $\partial : \text{Real}_{k-1}(X)$ , the visible boundary object

$$\partial_{\text{vis}}^{(k)}(X, R_k(n))$$

consists of the already exported unary traces of  $X$  against the recent layers, the binary comparison traces exported by the intervening layers, and all degenerate, reindexed, or transport faces generated transparently from those public traces.

The only face not included in this visible boundary is the direct remote comparison between the action of  $X$  and the newly exposed layer  $R_k(n)$ . The telescope of such missing comparison faces compatible with the endpoints of  $\partial_{\text{vis}}^{(k)}$  is written

$$\Gamma_{\text{miss}}^{(k)}(\partial).$$

For  $\gamma : \Gamma_{\text{miss}}^{(k)}(\partial)$ , the filler predicate

$$\text{Fill}_{\partial}^{(k)}(\gamma)$$

is the type of cubical fillers completing the open  $k$ -box whose visible faces are  $\partial_{\text{vis}}^{(k)}$  and whose missing face is  $\gamma$ . The structural horn-extension object is

$$H_k(\partial) := \sum_{\gamma : \Gamma_{\text{miss}}^{(k)}(\partial)} \text{Fill}_{\partial}^{(k)}(\gamma).$$

This is the central formal move of the paper. The marginal higher structural datum is a missing-face-plus-filler package. It is not a closed-boundary filler type. The missing face is chosen at the same time as the filler, and cubical composition computes the canonical choice.

**Definition 5.9** (Open-box extension type). Let  $\Gamma$  be a cubical context, let  $A : \Gamma, t : \mathbb{I} \vdash \mathcal{U}$  be a fibrant family, let  $\varphi$  be a visible-face cofibration, and suppose given compatible open-box data

$$u : (t : \mathbb{I}) \rightarrow \text{Partial}_{\varphi}(A(t))$$

and

$$u_0 : A(i0)[\varphi|_{t=i0} \mapsto u(i0)].$$

The open-box extension type is

$$\text{OpenExt}_\Gamma(A, \varphi, u, u_0) := \sum_{\ell: A(i1)[\varphi|_{t=i1} \mapsto u(i1)]} \text{Fill}(A, \varphi, u, u_0, \ell),$$

where  $\text{Fill}(A, \varphi, u, u_0, \ell)$  is the type of relative sections  $f : (t : \mathbb{I}) \rightarrow A(t)$  satisfying the base equation at  $i0$ , the lid equation at  $i1$ , and agreement with the partial boundary  $u(t)$  on  $\varphi$ .

The lid  $\ell$  is the missing remote face. The second component is the filler. Taking both components together is what turns the open-box extension problem into a contractible fiber.

**Lemma 5.10** (Horn-to-open-box adequacy). *For every derivation  $\Gamma \vdash H : \text{SHorn}_k(X)$  generated by the structural horn grammar, there are explicitly defined cubical open-box data*

$$\begin{aligned} A_H &: \Gamma, t : \mathbb{I} \vdash \mathcal{U}, & \varphi_H, \\ u_H &: (t : \mathbb{I}) \rightarrow \text{Partial}_{\varphi_H}(A_H(t)), \\ u_H^0 &: A_H(i0)[\varphi_H \mapsto u_H(i0)] \end{aligned}$$

and a canonical equivalence

$$\sum_{\gamma: \text{miss}(H)} \text{fill}(H)(\gamma) \simeq \text{OpenExt}_\Gamma(A_H, \varphi_H, u_H, u_H^0).$$

*Under this equivalence,  $\text{miss}(H)$  is identified with the compatible-lid subtype at  $i1$ ,  $\text{fill}(H)(\gamma)$  is identified with the corresponding relative PathP-style filler through the boundary-compatible family, and the construction is natural in cubical substitution.*

*Dependencies: structural horn grammar and its induction principle, the explicit open-box-extension type, and cubical reindexing of partial elements and compatible lids.*

*Proof.* By structural horn induction, Proposition 4.8.

For a remote-comparison horn, let  $A_H(t)$  be the family of comparison endpoints obtained by moving in the new interval direction from the already exported base face to the newly exposed remote layer. Let  $\varphi_H$  be the disjunction of the visible face formulas: the old unary action faces, the old binary comparison faces, and the transparent degeneracy and transport faces. The partial element  $u_H$  is the public boundary map assembled from those faces. The base element  $u_H^0$  is the compatible base comparison already present in  $\partial(H) : \text{Real}_{k-1}(X)$ . A missing remote comparison  $\gamma$  is precisely a compatible lid

$$\ell : A_H(i1)[\varphi_H \mapsto u_H(i1)],$$

and  $\text{fill}(H)(\gamma)$  is precisely the relative section from  $u_H^0$  to  $\ell$  agreeing with  $u_H$  on  $\varphi_H$ . This gives the displayed equivalence.

For a degenerate horn,  $A_H$ ,  $\varphi_H$ ,  $u_H$ , and  $u_H^0$  are obtained by applying the degeneracy map to the decoded open box of the lower horn. The compatible-lid subtype and relative filler type are the degenerate lid and degenerate filler types, so the equivalence is the lower equivalence transported along the degeneracy.

For a transported horn, the public path or equivalence transports the family  $A$ , the face formula, the partial boundary, and the base face. The decoded lid subtype is the transported missing-face type, and the decoded filler is the transported relative filler. The equivalence is therefore the transport of the induction hypothesis along these public equalities.

For a reindexed horn, the decoded tuple is

$$(A_{H'}, \varphi_{H'}, u_{H'}, u_{H'}^0)[\sigma].$$

The substitution action in Definition 4.7 gives  $\text{miss}(H'[\sigma]) = \text{miss}(H')[\sigma]$  and  $\text{fill}(H'[\sigma]) = \text{fill}(H')[\sigma]$ , while partial elements and subtype lids reindex functorially. Hence the open-box equivalence reindexes to the required equivalence. These clauses also prove the stated naturality in substitution.  $\square$

**Theorem 5.11** (Structural open-box extension contractibility). *For every structural open box  $(A, \varphi, u, u_0)$  generated by the sealing grammar in the fixed cubical core,*

$$\Gamma \vdash \text{isContr}(\text{OpenExt}_\Gamma(A, \varphi, u, u_0)).$$

*A center is computed by cubical composition and filling:*

$$(\text{hcomp}(A, \varphi, u, u_0), \text{hfill}(A, \varphi, u, u_0)).$$

*Moreover, for every cubical substitution  $\sigma : \Delta \rightarrow \Gamma$ , there is a canonical equivalence*

$$\text{OpenExt}_\Gamma(A, \varphi, u, u_0)[\sigma] \simeq \text{OpenExt}_\Delta(A[\sigma], \varphi[\sigma], u[\sigma], u_0[\sigma])$$

*that maps centers to centers, with contraction paths stable under substitution.*

*Dependencies: the explicit open-box-extension type, cubical composition and filling, and substitution stability of the generated boundary and lid subtype.*

*Proof.* The visible boundary is encoded as a cubical partial element. The compatibility equations on overlaps are exactly the well-typedness conditions for  $u$  and for the subtype element  $u_0$ . Homogeneous composition  $\text{hcomp}$  computes the compatible top lid, and cubical filling  $\text{hfill}$  supplies the relative section from the base to that lid. Thus  $\text{OpenExt}_\Gamma(A, \varphi, u, u_0)$  is the total type of endpoints together with dependent paths from the fixed base, i.e. a boundary-compatible path singleton rather than a fixed-lid filler space. This gives the displayed center.

Let  $(\ell, f)$  be any other lid-and-filler pair. To contract it to the canonical pair, apply cubical composition one dimension higher in the extension family. The boundary of this higher composition is given by the candidate filler  $f$ , the canonical filler, the visible partial boundary on  $\varphi$ , and the common base face at  $t = i0$ . The endpoint and overlap equations required for this higher box are precisely the filler laws for  $(\ell, f)$  and for the canonical extension. The resulting higher filler is a path from  $(\ell, f)$  to the canonical extension. Equivalently, this is the path-singleton contraction for the total space of a compatible lid together with a filler from the fixed base to that lid.

Uniformity of composition and filling under substitution gives the displayed reindexing equivalence and center-stability statement. No presentation quotient is used in this proof; it is a theorem about the theorem-facing cubical extension object.  $\square$

**Remark 5.12** (Fixed lids are a different problem). The preceding theorem contracts the total extension object

$$\sum_{\ell: A(i1)[\varphi|_{t=i1} \mapsto u(i1)]} \text{Fill}(A, \varphi, u, u_0, \ell).$$

It does not assert that, for an externally fixed lid  $\ell$ , the fiber

$$\text{Fill}(A, \varphi, u, u_0, \ell)$$

is contractible. Such a fiber can be empty when  $\ell$  is incompatible with the visible boundary, and even when inhabited it can carry the ambient higher homotopy of the family  $A$ . The singleton argument used by the theorem is the dependent-path singleton for “lid together with filler”, not a uniqueness theorem for fillers over arbitrary fixed closed data.

**Lemma 5.13** (Substitution stability of structural open boxes). *For every substitution  $\sigma : \Delta \rightarrow \Gamma$ , the reindexed tuple  $(A_k, \varphi_k, u_k, u_k^0)[\sigma]$  is the structural open box generated by the same sealing rule in context  $\Delta$ . The canonical center and contraction of Theorem 5.11 are stable under this reindexing.*

*Dependencies: the substitution action in the structural horn grammar and uniformity of the cubical composition and filling operations.*

*Proof.* Substitution acts structurally on interval expressions, face formulas, partial elements, subtypes, and the lower public boundary data from which  $(A_k, \varphi_k, u_k, u_k^0)$  is assembled. The cubical composition and filling operations are uniform under substitution. Therefore the reindexed open box is exactly the open box generated after first substituting the sealing rule, and the computed lid, filler, and contraction reindex to the corresponding computed lid, filler, and contraction in  $\Delta$ .  $\square$

**Remark 5.14** (Open boxes are not closed boundaries). Theorem 5.11 is an open-box extension theorem. It does not assert that the type of fillers for an arbitrary closed boundary is contractible. For a closed boundary, the filler fiber can carry genuine homotopy. The contractibility used here comes from the fact that the missing remote face is part of the extension data. In slogan form:

open-box extension type: contractible;

arbitrary closed-boundary filler type: not claimed.

This distinction is the reason the paper models marginal higher structural obligations as missing-face-plus-filler packages.

**Lemma 5.15** (Syntactic horn reduction for structural integration). *Let  $X$  be a foundational-core candidate sealed at stage  $n + 1$  in the fixed cubical extension calculus. For every  $k \geq 3$  and every boundary  $\partial : \text{Real}_{k-1}(X)$ , the additional data exposed by adjoining the remote layer  $L_{n-k+1}$  are canonically represented by the horn-extension fiber*

$$H_k(\partial) := \sum_{\gamma : \Gamma_{\text{miss}}^{(k)}(\partial)} \text{Fill}_{\partial}^{(k)}(\gamma).$$

*Every face of the box except the direct remote comparison face is already exported by unary and binary trace of the intervening sealed layers, together with transparent transport and degenerate faces.*

*Dependencies: arity-cell correspondence, factorization-complete export, the structural horn grammar, and the normal-form separation between payload and structural trace.*

*Proof.* By construction,  $\text{Real}_k(X)$  records only structural integration obligations induced by sealing  $X$  against opaque prior layers. Earlier layers therefore contribute only their exported payload and trace schemas; arbitrary higher-algebraic structure internal to those layers is invisible at this sealing boundary. Passing from depth  $k - 1$  to depth  $k$  exposes exactly one further remote layer, namely  $L_{n-k+1}$ .

The intervening sealed layers have already exported the unary bridges and binary comparison traces linking that remote layer to more recent history. Counting normalization treats the associated reindexings, degenerate faces, and transports as transparent. Thus all visible faces of the comparison box are fixed by lower public trace. The only new component is the direct remote comparison face  $\gamma$ , and once  $\gamma$  is chosen the remaining exact obligation is precisely the filler predicate. This is the displayed dependent sum.  $\square$



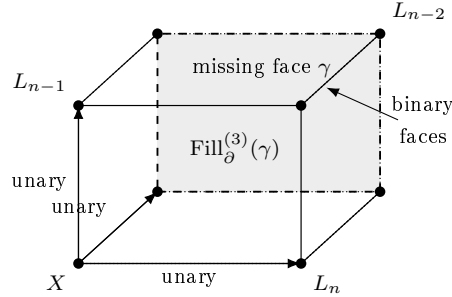


Figure 2: A depth-three structural open box. Unary action data and already exported binary comparison faces determine the boundary  $\partial$ ; the shaded back face is the missing remote comparison  $\gamma$ , and the interior cell is the filler computed by the cubical operations.

A concrete depth-three picture. This is the upper-bound branch of the running example: after the swap-respecting binary boundary is public, one more exported comparison route creates a marginal horn. In the abstract notation of the fixed calculus, the new layer  $X$  is compared with  $L_n$ ,  $L_{n-1}$ , and the newly exposed remote layer  $L_{n-2}$ . The visible faces include the unary bridges from  $X$  to the recent layers and the binary comparisons already exported by  $L_n$  and  $L_{n-1}$ . The missing face is the direct comparison between the two composites from  $X$  to  $L_{n-2}$ . The depth-three factor is therefore

$$H_3(\partial) = \sum_{\gamma: \Gamma_{\text{miss}}^{(3)}(\partial)} \text{Fill}_{\partial}^{(3)}(\gamma).$$

The term  $\gamma$  is not new payload. It is the missing remote face of the structural open box. Cubical composition computes a canonical  $\gamma$ , and filling packages the accompanying three-dimensional cell.

**5.3. Derivedness and minimal public signatures.** The open-box theorem is semantic. It says that the higher exact factor is contractible. The public-signature theorem is syntactic-accounting. It says that an explicit higher structural field in a surface presentation can be replaced by the cubically computed term and therefore removed from the primitive part of a  $\mu$ -minimal signature.

The bridge between the two levels is the computational replacement theorem from Section 3: when a presented field is accompanied by a uniform derivation from lower public boundary data, the presentation relation allows the field to be replaced by that derived term. The open-box theorem supplies exactly such a derivation for higher structural trace.

**Theorem 5.16** (Derived trace witnesses for higher structural horns). *Let  $h$  be a normalized structural horn package generated by the fixed cubical sealed-extension grammar, and suppose the historical arity of  $h$  is at least three. Then  $h$  carries a derived trace witness in the sense of Definition 4.10.*

*Dependencies: syntactic horn reduction, horn-to-open-box adequacy, total open-extension contractibility, substitution stability, and computational replacement on canonical trace presentations.*

*Proof.* Let  $\partial_h$  be the lower public boundary of  $h$ . By Lemma 5.15, the exact additional datum represented by  $h$  is a dependent sum

$$\sum_{\gamma: \Gamma_{\text{miss}}(\partial_h)} \text{Fill}_{\partial_h}(\gamma).$$

By Lemma 5.10, this sum is equivalent to the explicit structural open-box extension decoded from the horn grammar. The decoded box exposes the family  $A$ , the face formula  $\varphi$ , the partial boundary  $u$ , the base face  $u_0$ , the compatible lid subtype, and the filler family.

Theorem 5.11 gives a canonical center of this open-box extension, computed by cubical composition and filling. Transporting that center back along the horn-to-open-box equivalence gives the replacement term

$$t_h(\partial_h) : \sum_{\gamma: \Gamma_{\text{miss}}(\partial_h)} \text{Fill}_{\partial_h}(\gamma).$$

The typing and boundary equations are the endpoint and side equations of the decoded open box. Semantic preservation is the equivalence of Lemma 5.10. The allowed-dependency condition holds because the decoded box is assembled from  $\partial_h$  and the fixed cubical operations only. Substitution stability is Lemma 5.13. Finally, public-normalization compatibility is exactly the hypothesis consumed by the computational replacement theorem of Section 3: replacing the explicit horn field by the computed center gives a presentation of the same canonical trace signature in which the horn field is derived rather than primitive. These are precisely the components required by Definition 4.10.  $\square$

**Theorem 5.17** (Minimal-signature elimination above binary arity). *Let  $e$  be an admissible surface extension declaration over  $B_n$ . Then there exists a presentation-equivalent declaration  $e' \approx e$  whose normalized trace signature contains no irreducible primitive historical field of arity  $k \geq 3$ . Equivalently, every canonical  $\mu$ -minimal normalized public signature contains only unary and binary primitive structural trace fields.*

*More concretely, if a presented structural field of arity  $k \geq 3$  occurs in some surface declaration, then in the canonical trace-presentation interface it is transparently derived from its lower-arity boundary by a term synthesized from cubical composition, transport, and filling.*

*Dependencies: normal-form classification, derived trace witnesses for higher structural horns, and the computational replacement theorem.*

*Proof.* Fix a presentation of  $e$  and an apparent primitive structural trace field  $\phi$  of arity  $k \geq 3$  in its elaborated sealed export. By the normal-form classification of Section 4,  $\phi$  is a higher structural horn package rather than payload. By Theorem 5.16,  $\phi$  carries a **DerivedTrace** witness. Concretely, this witness supplies lower boundary data  $\partial$ , the replacement term  $t_\phi(\partial)$ , typing and boundary equations, semantic preservation, allowed-dependency data, substitution stability, and public-normalization compatibility.

The computational replacement rule of Section 3 then applies: a presented higher structural field equipped with such a uniform lower-boundary derivation is presentation-equivalent to the declaration in which the field is replaced by  $t_\phi(\partial)$  and classified as derived by the supplied witness. Repeating this replacement for every arity- $k \geq 3$  structural field terminates because the normalized signature is finite. The resulting presentation-equivalent declaration

$e'$  has no primitive structural field above binary arity. Since  $\mu$  counts only primitive trace fields in canonical normal form, every  $\mu$ -minimal signature has the same property.  $\square$

**Definition 5.18** (Primitive status and minimal opaque cost). After the replacement procedure of Theorem 5.17, an atomic trace schema in a normalized public signature is *derived* if it is synthesized by transparent normalization, by generic cubical structure, or by a supplied `DerivedTrace` witness. It is *primitive* if no such synthesis is available in the fixed normalization relation. For an admissible declaration  $e$ , let  $N_{\text{tr}}^{\text{prim}}(e)$  be the resulting primitive trace part. The minimal opaque public trace cost is

$$\mu(e) := \min\{|N_{\text{tr}}^{\text{prim}}(e')| \mid e' \approx e\}.$$

This is the formal version of the accounting notation introduced in Definition 2.7; in particular,  $\mu$  is attached to canonical normalized public signatures after witnessed replacement, not to raw declarations.

**Remark 5.19** (What is eliminated). Theorem 5.17 eliminates primitive public fields, not exact semantic factors. A higher horn factor may still occur in  $\text{Real}_k(X)$ , but after Theorem 5.11 it occurs as a contractible factor with a canonical center. The trace-cost invariant  $\mu$  does not count such a factor as primitive public trace.

**5.4. Exact stabilization above binary trace.** We can now prove the upper bound on obligation depth. The proof is conceptually simple: from depth two onward, each further exposure of remote history adjoins one structural open-box extension factor; each such factor is contractible; dependent sums with contractible fibers are equivalent to their bases.

**Lemma 5.20** (Kan contraction for horn fibers). *For every structural horn-extension fiber  $H_k(\partial)$  generated by the fixed cubical sealing grammar, the projection*

$$H_k(\partial) \rightarrow 1$$

*is contractible in the cubical metatheory. The center is the missing face and filler computed from  $\partial$  by the cubical operations.*

*Dependencies: horn-to-open-box adequacy and total open-extension contractibility.*

*Proof.* By Lemma 5.10,  $H_k(\partial)$  is the open-box extension type for the structural tuple  $(A_k, \varphi_k, u_k, u_k^0)$ . Applying Theorem 5.11 gives a canonical center and a contraction of the whole extension space. Presentation equivalence is not used to prove this contractibility; it is used only in Theorem 5.17 to classify the corresponding public field as derived.  $\square$

**Theorem 5.21** (Exact stabilization of realized structural obligations). *In the fixed cubical extension calculus, for every foundational-core candidate  $X$  and every  $k \geq 2$  the restriction map*

$$\text{res}_{2,k}(X) : \text{Real}_k(X) \longrightarrow \text{Real}_2(X)$$

*is an equivalence. Its inverse*

$$\text{ext}_{2,k}(X) : \text{Real}_2(X) \longrightarrow \text{Real}_k(X)$$

*is constructed by structural recursion over the generated obligations of arity greater than two, adjoining for each such obligation the canonical center of the decoded total open-extension object. The composites  $\text{res}_{2,k} \circ \text{ext}_{2,k}$  and  $\text{ext}_{2,k} \circ \text{res}_{2,k}$  are identified with the corresponding identities by the contraction paths of those open-extension fibers. Consequently  $d_{\text{obl}} \leq 2$ .*

*Dependencies: the formal definition of  $\text{Real}_k$ , arity-cell identification, syntactic horn reduction, horn-to-open-box adequacy, and total open-extension contractibility.*

*Proof.* The case  $k = 2$  is immediate. For  $k = 3$ , Lemmas 5.4 and 5.15 identify the marginal depth-three datum over each boundary  $\partial : \text{Real}_2(X)$  as the horn-extension fiber  $H_3(\partial)$ . Hence

$$\text{Real}_3(X) \simeq \sum_{\partial : \text{Real}_2(X)} H_3(\partial).$$

By Lemma 5.20, every fiber  $H_3(\partial)$  is contractible. A dependent sum over a family of contractible fibers is canonically equivalent to its base, so

$$\text{Real}_3(X) \simeq \text{Real}_2(X).$$

For  $k > 3$ , the same argument repeats. Passing from depth  $k - 1$  to depth  $k$  exposes one additional remote layer and adjoins one structural horn-extension fiber  $H_k(\partial)$  over  $\partial : \text{Real}_{k-1}(X)$ . Lemma 5.20 contracts that fiber, so

$$\text{Real}_k(X) \simeq \text{Real}_{k-1}(X) \quad (k \geq 3).$$

Iterating these equivalences yields  $\text{Real}_k(X) \simeq \text{Real}_2(X)$  for all  $k \geq 2$ . Written as maps, the forward map is the telescope projection  $\text{res}_{2,k}(X)$ . The inverse  $\text{ext}_{2,k}(X)$  starts with  $\partial : \text{Real}_2(X)$  and repeatedly adjoins the canonical center of each successive horn-extension fiber  $H_j(\partial_j)$ , for  $3 \leq j \leq k$ . The first round trip is judgmentally the identity on the retained depth-two fields. The second round trip is homotopic to the identity because each discarded higher component is connected to the canonical center by the contraction supplied by Lemma 5.20.  $\square$

**Corollary 5.22** (Contractible-factor decomposition). *For every foundational-core candidate  $X$  and every  $k \geq 2$ , there is a canonical equivalence*

$$\text{Real}_k(X) \simeq \text{Real}_2(X) \times C_k(X),$$

where  $C_2(X) := 1$  and, for  $k > 2$ ,  $C_k(X)$  is the iterated chain of canonical remote horn factors selected by the cubical compositions in the proof of Theorem 5.21. Each  $C_k(X)$  is canonically contractible.

*Dependencies: exact stabilization and Kan contraction for structural horn fibers.*

*Proof.* The proof of Theorem 5.21 trivializes each successive horn-extension fiber by choosing its canonical cubical center. Iterating those choices identifies the total higher part with a product of the depth-two boundary and the chain of chosen remote horn factors. Each factor is contractible by Lemma 5.20, hence the whole chain  $C_k(X)$  is contractible.  $\square$

The corollary expresses the exact semantic situation more vividly than the bare equivalence: above depth two there may be additional exact factors, but all of them are canonical and contractible. The primitive public signature sees only the depth-two part.

**5.5. From arity depth to a two-layer chronological window.** The exact-depth theorem bounds arity. It does not, by itself, identify which layers must be inspected in time. A primitive binary field could still compare a new layer with a remote old layer. The next theorem rules out such primitive remote binary fields for the fixed calculus, using the factorization-complete public exports introduced earlier.

**Theorem 5.23** (Recent-history factorization). *In the fixed cubical extension calculus, every binary trace field that survives in a  $\mu$ -minimal normalized public signature at stage  $n + 1$  factors through the previous two normalized signatures. More precisely, let  $X$  be sealed at stage  $n + 1$ . Once the unary and binary trace data of  $X$  against  $L_n$  and  $L_{n-1}$  have been fixed, every comparison between  $X$  and a remote layer  $L_{n-r}$ , for  $r \geq 2$ , is definitionally equal after normalization to a term obtained by iterating the already exported unary bridges and binary comparison traces of the intervening normalized signatures. Consequently no  $\mu$ -minimal public signature contains a primitive remote binary field.*

*Dependencies: factorization-complete export, the normal-form classification of structural trace, minimal-signature elimination, and the preceding exact stabilization theorem.*

*Proof.* Consider first the layer  $L_{n-2}$ . The sealed export of  $L_n$  contains the adjacent unary bridge to  $L_{n-1}$ , and the sealed export of  $L_{n-1}$  contains the adjacent bridge to  $L_{n-2}$ . The binary part of the trace exported by these layers records the comparison between the adjacent composite and the corresponding direct public route. Since exports are factorization-complete, the degenerate faces and transport faces needed to assemble the next comparison box are transparent public data.

When  $X$  is sealed against  $L_n$  and  $L_{n-1}$ , it supplies the unary comparisons to those two layers and the binary square comparing the two routes through them. Combining this data with the exported adjacent bridges determines all visible faces of an open three-box. The missing face is exactly the would-be direct comparison from  $X$  to  $L_{n-2}$ . Thus the remote comparison is not an independent primitive binary field; it is the lid of a structural horn-extension object. Cubical composition computes that lid, and cubical filling supplies the accompanying three-cell. By Theorem 5.17, the presented remote comparison is therefore replaced by the derived term in every  $\mu$ -minimal signature.

The argument iterates. Suppose the comparison from  $X$  to  $L_{n-r+1}$  has already been factored through recent exported signatures. The normalized export of  $L_{n-r+1}$  contains the adjacent bridge to  $L_{n-r}$ , the binary comparison trace for the composite through that bridge, and the transparent degenerate faces needed for the next open box. Combining these with the already factored comparison for  $X$  produces another structural horn-extension problem whose missing face is the comparison from  $X$  to  $L_{n-r}$ . Cubical composition and filling synthesize that face and filler. Hence every remote binary comparison is definitionally generated from the recent public exports and is not primitive in canonical minimal form.  $\square$

**Corollary 5.24** (Two-layer chronological window). *In the fixed cubical extension calculus, every sealing step at stage  $n + 1$  has chronological window size at most two. All primitive obligations are generated by the exported traces of  $L_n$  and  $L_{n-1}$ , while comparisons with layers  $L_{n-r}$  for  $r \geq 2$  factor through those two normalized signatures and are synthesized by iterated cubical filling.*

*Dependencies: exact stabilization and recent-history factorization.*

*Proof.* Theorem 5.21 shows exact arity stabilization at depth two, while Theorem 5.23 shows that surviving binary primitive fields can be chosen to factor through the two most recent normalized signatures. Therefore deeper history contributes only derived fillers and already mediated boundary data, not primitive public trace fields outside the recent two-layer window.  $\square$

The recurrence theorem in Section 6 starts only after this point. It counts the active basis inside the two-layer chronological window under an additional full-coupling hypothesis.

**5.6. The lower bound: univalence forces binary trace.** It remains to show that the upper bound is exact. The obstruction is small but decisive. A univalent interface that exports the two-point type also exports the path corresponding to its nontrivial automorphism. A sealed action must respect that path. Objectwise unary data alone cannot express this naturality condition. This is the lower-bound branch of the running example from the introduction.

**Definition 5.25** (Unary and binary fragments for the swap test). For a candidate  $X$  over the two-point interface, the unary fragment  $\text{Real}_1(X)$  records only the objectwise action field

$$F_2 : 2 \rightarrow 2.$$

The binary fragment  $\text{Real}_2(X)$  extends this record by the comparison field forced by the exported univalent swap path:

$$\alpha_{\text{swap}} : \text{transport}_{\lambda A.A \rightarrow A}(p_{\text{swap}}, F_2) = F_2.$$

The map  $\text{res}_{1,2} : \text{Real}_2(X) \rightarrow \text{Real}_1(X)$  forgets  $\alpha_{\text{swap}}$ . Thus a unary-accepted candidate is an inhabitant of  $\text{Real}_1(X)$ , while binary acceptance asks for a point in the fiber of  $\text{res}_{1,2}$  over that unary action.

**Theorem 5.26** (Explicit binary sealing obstruction). *Let the active interface of a univalent foundation export the two-point type 2 and the univalent path*

$$p_{\text{swap}} := \text{ua}(\text{swap}) : 2 = 2.$$

*Then there is a candidate  $X_{\text{const}}$  and an element*

$$x_1 : \text{Real}_1(X_{\text{const}})$$

*such that the fiber*

$$\sum_{x_2 : \text{Real}_2(X_{\text{const}})} \text{res}_{1,2}(x_2) = x_1$$

*is empty. Equivalently, the unary fragment admits a raw objectwise action that the binary fragment rejects. Consequently depth-one data are insufficient in general, and  $d_{\text{obl}} \geq 2$ .*

*Dependencies: the univalent two-point interface, transport in the endomorphism family, and the distinction between unary objectwise action and binary sealing naturality.*

*Proof.* Consider an active-interface package consisting of an endomap on the exported two-point type:

$$X_{\text{act}} := (F_2 : 2 \rightarrow 2).$$

The unary candidate chooses the constant-left map

$$F_2 := \lambda\_ . \text{left}.$$

This is a well-formed objectwise clause, so it determines the point  $x_1 : \text{Real}_1(X_{\text{const}})$ . But sealing a globally acting structural extension requires compatibility with exported paths. Since the old public interface exports  $p_{\text{swap}}$ , the binary naturality obligation in  $\text{Real}_2(X_{\text{const}})$  is

$$\alpha_{\text{swap}} : \text{transport}_{\lambda A.A \rightarrow A}(p_{\text{swap}}, F_2) = F_2.$$

By univalence, transport in the endomorphism family along  $p_{\text{swap}}$  computes by conjugation with the swap equivalence. Therefore

$$\text{transport}_{\lambda A.A \rightarrow A}(p_{\text{swap}}, F_2) = \text{swap} \circ F_2 \circ \text{swap}^{-1} = \lambda\_ . \text{right}.$$

Thus  $\alpha_{\text{swap}}$  would be an equality

$$(\lambda\_.\text{right}) = (\lambda\_.\text{left}),$$

which is impossible because evaluating at **left** would force **right** = **left**.

There is also a positive control: if  $F_2$  is the identity map, conjugation along swap returns the identity map, and the binary obligation is inhabited. The failure of the constant-left map is therefore not a defect of the formulation; it is the type checker detecting an objectwise action that does not respect exported path structure. Hence the fiber of  $\text{res}_{1,2}$  over the constant-left unary point is empty:  $\text{Real}_1$  can be inhabited while the corresponding  $\text{Real}_2$  extension is impossible. Thus  $\text{Real}_1$  and  $\text{Real}_2$  are not canonically equivalent in general, and the obligation depth cannot be one.  $\square$

**Remark 5.27** (Adjunctions as an explanatory analogue). Adjunctions exhibit the same binary pattern categorically. Once the object action and the unit/counit data have been fixed, the triangle identities compare composites already determined at the unary level. The paper uses this only as an analogy. The internal lower bound for the fixed cubical calculus is the swap obstruction of Theorem 5.26.

**5.7. Exact public structural depth two.** We can now combine the upper and lower bounds.

**Corollary 5.28** (Fixed-calculus exact structural integration depth two). *In the fixed cubical sealed-extension calculus, admissible sealed structural extensions have exact structural integration depth two:*

$$d_{\text{obl}} = 2.$$

*Moreover every  $\mu$ -minimal normalized public signature contains no primitive structural trace field of historical arity greater than two, and every sealing step has a two-layer chronological window under the factorization-complete export hypothesis.*

*Dependencies: exact stabilization for the upper bound, minimal-signature elimination for the public-signature statement, recent-history factorization for the chronological window, and the swap obstruction for the lower bound.*

*Proof.* Theorem 5.21 gives the upper bound  $d_{\text{obl}} \leq 2$  by showing  $\text{Real}_k(X) \simeq \text{Real}_2(X)$  for all  $k \geq 2$ . Theorem 5.17 turns the same cubical horn computation into the public-signature statement  $d_\mu \leq 2$ . Theorem 5.23 and Corollary 5.24 give the separate chronological factorization needed for the later recurrence. Finally, Theorem 5.26 gives the lower bound  $d_{\text{obl}} \geq 2$  by exhibiting a genuine binary sealing obstruction in the univalent setting. Hence the exact obligation depth is two.  $\square$

The result should be read with the distinctions of Remark 5.2 in mind. Higher structural exact factors do not vanish from the semantics; they stabilize as contractible open-box extension factors. What disappears from the primitive public interface is the need to present those factors as irreducible fields. This is the semantic content behind the paper's slogan:

binary trace is necessary; higher structural trace is cubically payable.

## 6. CONDITIONAL COUNTING CONSEQUENCES

This section records only the counting consequence used by the paper. The recurrence is not part of the exact-depth theorem itself; it is a conditional accounting corollary for presentations satisfying the full-coupling hypothesis.

For a sealed layer  $L_n$ , let  $\kappa_n$  be the payload size of the new sealed layer, let  $\mu_n$  be the primitive public trace cost of its normalized opaque signature, and let  $b_n$  be the active primitive basis size after sealing  $L_n$ . A sparse presentation pays only for the primitive basis classes that occur in its footprint. A full-coupling presentation pays against the entire active basis exposed by the depth window. Thus sparse footprint is a local dependency notion, while full coupling is a maximal accounting envelope.  $\kappa_n$  is a new basis summand, not itself structural trace;  $\mu_n$  is counted only after derived replacement; missing earlier indices are read as zero.

| Symbol     | Meaning                                                  | Counted as primitive trace?       |
|------------|----------------------------------------------------------|-----------------------------------|
| $\kappa_n$ | payload basis exported by $L_n$                          | no, payload basis                 |
| $\mu_n$    | primitive trace cost after replacement and normalization | yes                               |
| $b_n$      | active primitive basis available to the next layer       | only through selected trace sites |
| $d$        | chronological window supplied before accounting starts   | no                                |

**Definition 6.1** (Layerwise quantities and active basis). Under a depth- $d$  window, the active basis size is

$$b_n = \kappa_n + \sum_{i=1}^d \mu_{n-i}$$

with missing earlier indices interpreted as zero. A candidate is *fully coupled* when each primitive basis class in this active window is used by the exported trace footprint of the layer.

**Corollary 6.2** (Conditional full-coupling accounting recurrence under a depth- $d$  window). *Assume the fixed-calculus adequacy bridge, the bundled per-site counting convention, and the full-coupling hypothesis. If the active primitive basis after sealing layer  $n$  is obtained from the new payload together with the primitive trace exported by the preceding  $d$  active layers, then*

$$\mu_n = \kappa_n + \mu_{n-1} + \cdots + \mu_{n-d}.$$

*Proof.* By the active-window theorem, primitive support for sealed structural trace factors through the displayed depth- $d$  basis. Full coupling says that every class in that basis is used, and  $\mu$ -minimality counts one primitive public trace representative for each primitive class. The displayed sum is therefore exactly the active basis cardinality.  $\square$

**Corollary 6.3** (Depth-two full-coupling accounting specialization). *In the fixed cubical sealed-extension calculus, the exact-depth theorem gives the depth-two window. Under full coupling,*

$$\mu_n = \kappa_n + \mu_{n-1} + \mu_{n-2}.$$

*Under the additional constant-payload and empty-history assumptions, the shifted sequence  $\mu_n + 2c$  satisfies the Fibonacci recurrence.*



## 7. MECHANIZATION AND TRUSTED BOUNDARY

The artifact checks the theorem-facing fixed-calculus stack and records the boundary between checked Agda statements, abstract interfaces, and paper-level adequacy assumptions. The detailed artifact guide, recurrence tables, and case-study walkthroughs are intentionally kept out of the LMCS paper and belong to the archived release.

The paper and the machine-readable theorem map use the following four status phrases. A claim is *fully mechanized* when the named Agda modules check the claim directly for the fixed object under discussion, with no local postulates or local primitive declarations beyond the trusted cubical library. It is *mechanized for an abstract interface* when Agda checks a precise interface theorem, but an instantiation or adequacy argument is a separate input. It is *conditional on adequacy* when checked components are present but the paper-level claim depends on the stated bridge from surface presentations or semantic wrappers to  $C_{\text{ext}}$ . It is *paper-only* when the item is an expository, release, or future-transfer claim rather than a theorem checked by the artifact.

For this submission, `paper-map.yaml` remains the machine-readable theorem map for the full repository paper source. The LMCS file deliberately compresses several labels into the table below; the bridge from those compressed LMCS references back to the checked Agda surface is documented in the repository theorem index rather than by retargeting the existing checker.

| Paper claim                                                             | LMCS reference               | Status                               | Artifact surface                                                          | Trusted input or condition                                                      |
|-------------------------------------------------------------------------|------------------------------|--------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Fixed calculus and generated obligations                                | Section 2, Definition 2.1    | fully mechanized                     | raw structural syntax/typing and normalization bridge modules             | fixed raw syntax and Cubical library                                            |
| ObSig <sub>k</sub> , Real <sub>k</sub> , restriction, and stabilization | Definition 5.1, Theorem 5.21 | mechanized for an abstract interface | upper-bound and exact-depth modules                                       | theorem-facing realization and horn interfaces                                  |
| Horn-to-open-box decoding                                               | Lemma 5.10                   | mechanized for an abstract interface | structural horn decoding and open-box bridge modules                      | generated horn grammar; raw/surface use needs the bridge                        |
| Total extension tractability                                            | open-Theorem 5.11            | mechanized for an abstract interface | cubical open-box extension, contraction, and substitution modules         | cubical <b>hcomp</b> , <b>hfill</b> , and Partial/Sub; not fixed-lid uniqueness |
| Derived-trace placement $\mu$ -minimal elimination                      | re-Theorems 5.16 and 5.17    | conditional on adequacy              | normalization-derived, replacement, and computational-replacement modules | canonical trace presentations and fixed raw bridge                              |
| Binary lower bound                                                      | Theorem 5.26                 | fully mechanized                     | adjunction-barrier, depth-lower-bound, and exact-depth modules            | univalence and the two-point swap interface                                     |
| Exact public depth-two conclusion                                       | Corollary 5.28               | conditional on adequacy              | exact-depth wrappers plus replacement and chronology modules              | adequacy bridge for public signatures; factorization for the window claim       |
| Counting recurrence                                                     | recur-Section 6              | conditional on adequacy              | sparse/full-coupling recurrence and semantic scaling modules              | full-coupling assumptions and the adequacy reading of $\mu$                     |
| Concrete SMod surface instance                                          | sur-Section 3                | fully mechanized                     | modal surface modules and smoke tests                                     | this concrete surface calculus only                                             |
| Broad transfer to arbitrary cubical languages                           | non-claim                    | paper-only                           | no theorem in this artifact                                               | external parser/elaborator and semantic adequacy theorem required               |

The trusted base consists of the Agda kernel with cubical support, the pinned Cubical Agda library used by the repository, the fixed raw extension-calculus records that define  $C_{\text{ext}}$ , and the small arithmetic layer used by the counting corollary. The theorem map audits the theorem-facing import closure for local postulates and local primitive declarations; imported cubical primitives remain part of the trusted library base. A checked module therefore supports exactly the row in which it appears. It does not by itself claim that every surface language or every cubical foundation instantiates the fixed interface.

The archived artifact contains the paper source, Agda modules, theorem map, audit scripts, and case-study fixtures. Reviewers can run the aggregate check from the release root; it typechecks the theorem-facing aggregate, verifies the paper-to-Agda map, and audits the theorem-facing import closure for local postulates and primitive declarations. The fixture names are intentionally diagnostic: **universe-extension-summary** and **global-modality-summary** are full-coupling examples, **sparse-datatype-summary** is sparse local growth, **transparent-lemma-summary** is a zero-footprint transparent addition, and **higher-payload-summary** records higher user payload that is not structural trace. The development repository is not the archival citation; the LMCS submission should cite the

Zenodo release DOI once reserved. The final archival tag, DOI, and hosted artifact-check URL are release metadata, not additional theorem claims.

## 8. CONCLUSION

This paper proves an exact depth-two theorem for the fixed cubical sealed-extension calculus  $\mathcal{C}_{\text{ext}}$ . Unary trace is insufficient: univalence exposes public comparisons, and the two-point swap gives a concrete binary sealing obstruction. Trace above binary arity is still present in the exact semantic object, but it is represented as a cubical open-box extension factor and is contractible over its lower boundary. Computational replacement therefore removes such fields from  $\mu$ -minimal normalized public signatures.

The binary boundary is exact because it is where public comparison first enters. Objectwise action is unary, but naturality along a public equivalence is binary. In the running two-point example, the constant-left action exposes the lower bound by failing this comparison, while the identity action shows the positive branch on which higher questions can be asked. After that binary boundary is present, sealing-generated higher coherence data has the shape cubical composition is designed to fill: lower faces are public, the missing face is computed, and the filler is canonical up to the contractibility proved in the fixed interface.

The result is deliberately scoped. It does not eliminate higher paths, forbid higher payload, or claim that arbitrary Cubical Agda developments satisfy the same accounting discipline. The recurrence in Section 6 is conditional on full coupling, and transfer to other foundations requires the stated adequacy and normalization interface. The mechanization records this trusted boundary rather than hiding it.

Future work is to instantiate the interface for broader cubical foundations, separate the full-coupling envelope from realistic sparse-growth models, and turn the current adequacy bridges into reusable criteria for structural extensions.

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